MICROSOFT 365

CHEAT SHEETS

Word, Excel, PowerPoint & Outlook — Shortcuts, Formulas and Automation

Ribbon & KeyTips · OneDrive · Templates · Styles · Long Documents · Mail Merge · Track Changes · Excel Tables · XLOOKUP · Dynamic Arrays · LET & LAMBDA · GROUPBY · PivotTables · Power Query · Slide Master · Morph · Presenter View · Outlook Rules · Quick Steps · Teams · VBA · Office Scripts · Copilot

39 REFERENCE SHEETS

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Sheet 01

Introduction to Microsoft 365

M365 vs Office 2024 · desktop · web · mobile · Microsoft 365 Copilot app · .docx .xlsx .pptx

ONE SUBSCRIPTION, THREE WAYS IN

Microsoft 365 is the subscription that bundles Word, Excel, PowerPoint, Outlook, OneDrive and Teams. Each app comes in three flavours: the **desktop apps** (Windows and Mac, the most complete), the **web apps** in the browser (free with a Microsoft account, fewer features), and **mobile apps**. **Office 2024** is the one-time-purchase version: same core apps, no new features after release and no Copilot.

NOTES

- Start page: `m365.cloud.microsoft` (older `office.com` links redirect)
- The **Microsoft 365 Copilot app** is the launcher on Windows / mobile: recent files, apps, Copilot chat
- Windows desktop has the most features; Mac lags on VBA, Power Query editing and some add-ins
- Files saved to **OneDrive** or **SharePoint** get AutoSave, sharing and version history
- Free web apps need only a Microsoft account; desktop apps need a paid plan or Office 2024

OPEN THINGS FAST

```
word.new  excel.new  powerpoint.new   # new blank file in the browser (OneDrive)
ppt.new   xls.new    doc.new            # short aliases for the same

⌘N        / Ctrl+N     # new blank file (desktop)
⌘O        / Ctrl+O     # open
⌘S        / Ctrl+S     # save (AutoSave on: saves a version)
⌘W        / Ctrl+W     # close window
⌘P        / Ctrl+P     # print / export PDF
⌘,        / File › Options   # preferences / options
⌘Q        / Alt+F4     # quit
```

CODE EXPLAINED

→ `word.new` and friends work in any browser address bar and create the file in your OneDrive

→ Shortcuts are listed **macOS first**, then **Windows**, on every sheet in this set

→ Mac preferences live under the app menu (`⌘,`); Windows uses **File › Options**

→ Web apps reuse most Windows shortcuts, but some browser shortcuts win (`Ctrl+N` opens a browser window)

OTHER SUITES

⇄ **Google Workspace** — `docs.new`, `sheets.new`, `slides.new`; everything lives in Drive

⇄ **Apple iWork** — Pages, Numbers, Keynote are free on Mac / iPad; icloud.com for the web versions

PLATFORM DIFFERENCES

- **Windows** — full feature set, VBA, Power Query, all add-ins
- **Mac** — near parity; some features arrive later and VBA has gaps
- **Web** — free, co-authoring first, no VBA macros, fewer formats

WHICH VERSION AM I ON?

EDITION	WHAT YOU GET
`Microsoft 365 Personal / Family`	Desktop + web + mobile, 1 TB OneDrive each, monthly updates
`Microsoft 365 Business / E plans`	Same apps plus Teams, SharePoint, admin, Exchange mail
`Office 2024 (Home / LTSC)`	One-time purchase, desktop only, no feature updates
`Free web apps`	Browser only, Microsoft account, 5 GB OneDrive
`Copilot add-on`	AI in the apps (some features included in Personal / Family)

FILE FORMATS

EXT	MEANING
`.docx .xlsx .pptx`	Standard files (no macros)
`.docm .xlsm .pptm`	Macro-enabled versions
`.dotx .xltx .potx`	Templates
`.xlsb`	Binary workbook: smaller, faster for big data
`.csv`	Plain text; loses formatting, formulas and extra sheets

⚠ COMMON MISTAKES

- Saving to the local disk and then wondering why AutoSave and version history are greyed out
- Assuming a feature exists on the web or Mac because you saw it on Windows; check the Platform box
- Saving as `.csv` and losing every other sheet and all formulas without noticing

✓ TIPS & BEST PRACTICES

- Type `excel.new` in the address bar instead of hunting for the New button
- Check **File › Account** (Windows) or **Help › Check for Updates** (Mac) to see your version
- Pin the apps you use to the taskbar / Dock; open recent files from their jump list

QUICK SUMMARY

- `M365` — subscription, monthly features
- `Office 2024` — one-time, frozen
- `*.new` — instant blank file in the browser
- `.xlsm` — the only Excel format that keeps macros

Sheet 02

Ribbon, Search & KeyTips

Simplified / Classic ribbon · Quick Access Toolbar · Alt KeyTips · Microsoft Search · customize ribbon · ⌘⌥R / Ctrl+F1

STOP HUNTING THROUGH TABS

Every command lives somewhere on the **ribbon**, but the fastest way to a command you can't remember is the **Search box** at the top of the window: type what you want and press Enter. On Windows, pressing **Alt** shows **KeyTips**, letters for every tab and button, so any ribbon command becomes a keyboard sequence. The **Quick Access Toolbar** (QAT) holds the handful of commands you use all day.

NOTES

- Search box (Windows): `Alt+Q`; it finds commands, help and recent files
- KeyTips: press and release `Alt`, then the letters, e.g. `Alt H B A` = all borders
- Number keys run QAT buttons on Windows: `Alt+1`, `Alt+2` …
- Simplified ribbon (one line) is the default on the web; switch with the ribbon's dropdown
- Mac has no KeyTips; use the Help menu search (`⇧⌘/`) or the app's own search box

AUTOMATE IT

- Add your top 5 commands to the QAT and run them with `Alt+1…5` on Windows
- Export ribbon + QAT customizations (File › Options › Customize Ribbon › Import/Export) to reuse on a new PC

RIBBON & KEYTIP SEQUENCES

```
⌘⌥R       / Ctrl+F1        # collapse / expand the ribbon
⇧⌘/       / Alt+Q          # search commands (Help menu on Mac)
—         / Alt  then F10  # show KeyTips (Windows)

# KeyTip sequences (Windows) — press one after another
Alt H B A      # Home › Borders › All Borders
Alt H O I      # Home › Format › AutoFit column width (Excel)
Alt H V V      # Paste values (Excel)
Alt N T        # Insert › Table
Alt W F F      # View › Freeze Panes (Excel)
Alt 1          # first Quick Access Toolbar button
```

CODE EXPLAINED

- → KeyTips are **sequences**, not chords: tap `Alt`, release, then type each letter
- → Old Office 2003 menu sequences (`Alt E S V` for paste values) still work in Excel
- → Right-click any ribbon button › **Add to Quick Access Toolbar**
- → Collapsing the ribbon gives you more rows; click a tab to peek at it

OTHER SUITES

- ⇄ **Google Workspace** — search the menus with `⌥/` / `Alt+/`; menus instead of a ribbon
- ⇄ **Apple iWork** — toolbar + Format sidebar; customize via View › Customize Toolbar

PLATFORM DIFFERENCES

- ▣ **Windows** — KeyTips, QAT number shortcuts
- ▣ **Mac** — no KeyTips; Help menu search and ribbon tabs only
- ▣ **Web** — KeyTips work with `Alt+Win` in some browsers; search box is the main route

WHERE THINGS LIVE

TAB	GO HERE FOR
`Home`	Clipboard, fonts, paragraph / alignment, styles
`Insert`	Tables, pictures, charts, links, symbols
`Layout / Design`	Margins, page setup, themes
`Review`	Spelling, comments, Track Changes, protect
`View`	Zoom, panes, gridlines, macros
`Developer`	Macros, VBA editor, form controls (turn on first)

CUSTOMIZE

TASK	WHERE
`Add QAT button`	Right-click a button › Add to QAT
`Show QAT below ribbon`	QAT dropdown › Show Below the Ribbon
`New ribbon tab / group`	File › Options › Customize Ribbon
`Turn on Developer tab`	Customize Ribbon › tick Developer
`Mac ribbon prefs`	Word / Excel › Settings › Ribbon & Toolbar

⚠ COMMON MISTAKES

- Holding `Alt` down while typing KeyTip letters; it must be pressed and released
- Customizing the ribbon on one PC and expecting it to follow you; export the file
- Hunting through tabs for a command you could type into the Search box in two seconds

✓ TIPS & BEST PRACTICES

- Learn the KeyTips for the 3 commands you click most; they never change between versions
- Turn on the **Developer** tab once; macros and form controls live there
- On Mac, add menu shortcuts for anything in System Settings › Keyboard › Keyboard Shortcuts › App Shortcuts

QUICK SUMMARY

- `Alt` — KeyTips (Windows)
- `Alt+Q` / `⇧⌘/` — search commands
- `QAT` — your favourites, `Alt+1…`
- `⌘⌥R` / `Ctrl+F1` — hide the ribbon

Sheet 03

OneDrive, AutoSave & Sharing

AutoSave · Version History · share links · co-authoring · @mentions · Files On-Demand · SharePoint

SAVE ONCE, NEVER AGAIN

When a file lives in **OneDrive** (personal) or **SharePoint / Teams** (work), **AutoSave** saves every change, **Version History** keeps older copies, and several people can edit at the same time (**co-authoring**). **Sharing** sends a link instead of an attachment, so everyone works on one copy. **Files On-Demand** shows all your cloud files in Finder / File Explorer without downloading them.

NOTES

- AutoSave toggle sits top-left of the window; it's off for local files
- Version History: **File › Info › Version History** (Win) or the file name dropdown
- `@mention` someone in a comment to email them a link to that spot
- Link types: **Anyone**, **People in your organization**, **Specific people**
- Co-authors' cursors show as coloured flags; their changes appear within seconds

SHARE A FILE THE RIGHT WAY

1. Save the file to OneDrive or SharePoint (File › Save As › pick a cloud location)
2. Click **Share** (top right) and choose who the link works for
3. Set **Can edit** or **Can view**; add an expiry date or block download if needed
4. Type names to send, or **Copy link** to paste into chat / email
5. Use **Manage Access** later to see and remove everyone who has a link
6. Restore a bad edit from **Version History** › Restore

WHERE TO FIND IT

TASK	WHERE / SHORTCUT
Share	Share button (top right)
Version History	File › Info › Version History
New comment	⌥⌘A / Ctrl+Alt+M
Who is editing	Avatars, top right
Manage links	Share › ··· › Manage Access
Free up space	Right-click › Free up space (On-Demand)

OTHER SUITES

- ⇄ **Google Workspace** — everything autosaves in Drive; Share button and File › Version history the same way
- ⇄ **Apple iWork** — iCloud Drive + Share › Collaborate; File › Revert To › Browse All Versions

PLATFORM DIFFERENCES

- ▣ **Windows** — OneDrive in File Explorer, Files On-Demand
- ▣ **Mac** — OneDrive app in Finder (File Provider); same on-demand idea
- ▣ **Web** — always AutoSave; version history from the file name menu

LINK SETTINGS

SETTING	USE WHEN
Anyone with the link	Public handouts; anyone can forward it
People in your org	Internal docs; needs sign-in
Specific people	Sensitive files; only named addresses
Can view / Can edit	Default to view; grant edit on purpose
Expiry / block download	Anyone links on paid plans

ATTACHMENT VS LINK

	ATTACHMENT	LINK
Copies	One per recipient	One shared file
Latest version	Whatever you sent	Always current
Revoke access	Impossible	Manage Access
Size limit	Mail limit (~20–35 MB)	None practical

⚠ COMMON MISTAKES

- Emailing attachments back and forth and merging edits by hand; share one link instead
- Using **Anyone** links for sensitive files; they work for whoever the link is forwarded to
- Turning AutoSave off to "experiment" and forgetting it; use **Save a Copy** first instead

✓ TIPS & BEST PRACTICES

- Name important points in history by saving a copy (`File › Save a Copy`) before big changes
- Use `@mentions` in comments to assign tasks; Word and Excel can turn them into tasks
- Mark folders **Always keep on this device** when you'll be offline

QUICK SUMMARY

- `AutoSave` — cloud files only
- `Version History` — restore any earlier copy
- `Share` — one link, set view / edit
- `@mention` — notify a person in a comment

Sheet 04

Templates & Themes

File › New · personal templates folder · .dotx .xltx .potx · document themes · Normal.dotm · Book.xltx

START FROM SOMETHING GOOD

A **template** is a file you start from instead of a blank page: styles, layouts, headers, formulas and placeholder text already set up. Save your own as `.dotx`, `.xltx` or `.potx` and they appear under **File › New › Personal**. A **theme** is a set of colours, fonts and effects that styles, charts and SmartArt follow, so changing the theme restyles the whole file at once.

NOTES

- Personal templates folder (Win): set in **File › Options › Save › Default personal templates location**
- Mac: templates saved via **File › Save as Template** go to the My Templates folder automatically
- `Normal.dotm` is Word's blank-document template; changing its styles changes every new doc
- Excel uses `Book.xltx` in the **XLSTART** folder as the default new workbook if it exists
- Design › Themes (Word), Page Layout › Themes (Excel), Design (PowerPoint)

AUTOMATE IT

- Save a styled file as a template once instead of copying last month's file and deleting content
- Organization templates in SharePoint show up under File › New for everyone

MAKE YOUR OWN TEMPLATE

1. Build a file with your styles, theme, headers / footers, logo and placeholder text
2. Delete real content but keep placeholders like "[Client name]"
3. **File › Save As** › file type **Word / Excel / PowerPoint Template**
4. Windows: save into your personal templates folder (set the folder path first)
5. Next time: **File › New › Personal** (Win) or **File › New from Template** (Mac)
6. To change it later, open the template file itself (right-click › Open, not double-click)

WHERE TO FIND IT

TASK	WHERE
Template gallery	File › New
Save as template	File › Save As › .dotx / .xltx / .potx
Change theme	Design › Themes
Theme colours / fonts	Design › Colors / Fonts › Customize
Set as default	Design › Set as Default (Word)
Save theme	Themes › Save Current Theme (.thmx)

OTHER SUITES

⇄ **Google Workspace** — Template gallery; custom org templates on Workspace; `/copy` share links

⇄ **Apple iWork** — File › Save as Template adds it to the template chooser

PLATFORM DIFFERENCES

- **Windows** — personal templates folder must be set before Personal appears
- **Mac** — My Templates folder is automatic
- **Web** — use built-in templates; open a template from OneDrive and save a copy

TEMPLATE FILES

TYPE	APP	HOLDS
.dotx	Word	Styles, headers, Quick Parts
.dotm	Word	Same plus macros (Normal.dotm)
.xltx	Excel	Sheets, formulas, formats, validation
.potx	PowerPoint	Slide masters, layouts, theme
.thmx	All three	Colours + fonts + effects only

WHAT A THEME CONTROLS

PART	AFFECTS
Colors (12 slots)	Styles, charts, tables, SmartArt
Fonts (heading + body)	Every style using +Headings / +Body
Effects	Shape shadows, bevels, lines
Style set (Word)	Spacing and look of built-in styles

⚠ COMMON MISTAKES

- Copying an old report and deleting content; hidden comments and tracked changes come along
- Double-clicking a template to edit it; that opens a new copy, not the template
- Hard-coding colours instead of theme colours, so a theme change misses half the file

✓ TIPS & BEST PRACTICES

- Use theme fonts (+Headings / +Body) in styles so the font changes in one place
- Keep one master template per document type and version it in OneDrive
- In Excel templates, lock formula cells and protect the sheet before saving

QUICK SUMMARY

- `.dotx / .xltx / .potx` — templates
- `File › New › Personal` — your templates
- `Theme` — colours + fonts + effects
- `Normal.dotm` — Word's default template

Shortcuts That Work Everywhere

LEARN ONCE, USE IN EVERY APP

Word, Excel, PowerPoint and Outlook share one set of core shortcuts. The biggest time-savers are **Paste Special** (paste as plain text or values), **Repeat** (do the last action again), **Format Painter** (copy formatting) and **Undo** history. Learn these once and they work in every app, on the web too in most cases.

NOTES

- `Repeat` (`⌘Y` / `F4`) repeats formatting, inserts, deletes: anything you just did
- Double-click **Format Painter** to paint many places; `Esc` to stop
- **Paste as text**: Paste Options › Keep Text Only, or `⌥⇧⌘V` (Mac Word) / `Ctrl+Shift+V` (web)
- Dictate: **Home › Dictate**, or `Win+H` for Windows system dictation
- `Shift+F3` cycles UPPER / lower / Title case in Word and PowerPoint

SHARED SHORTCUTS

```
⌘C ⌘X ⌘V      / Ctrl+C X V      # copy, cut, paste
^⌘V           / Ctrl+Alt+V      # Paste Special dialog
⌘Z  ⌘Y        / Ctrl+Z  Ctrl+Y  # undo / redo or repeat
⌘Y            / F4              # repeat last action
⇧⌘C ⇧⌘V       / Ctrl+Shift+C V  # copy / paste formatting (Word, PPT)
⌘B ⌘I ⌘U      / Ctrl+B I U      # bold, italic, underline
⌘K            / Ctrl+K          # insert link
⌘F            / Ctrl+F          # find
^H / ⇧⌘H      / Ctrl+H          # replace
F7            / F7              # spelling & grammar
⇧F3           / Shift+F3        # change case (Word, PPT)
⇧⌘S           / F12             # Save As
```

CODE EXPLAINED

- → On Windows, `Ctrl+Y` redoes after an undo and repeats otherwise; `F4` is the same
- → Mac function keys may need `fn` (`fn+F4`) unless you set "Use F1, F2… as standard keys"
- → Copy / paste formatting (`⇧⌘C`) is the keyboard version of Format Painter
- → Excel's Paste Special has its own letters: `V` values, `T` formats, `E` transpose

OTHER SUITES

- ⇄ **Google Workspace** — `⌘⇧V` / `Ctrl+Shift+V` paste without formatting; paint format button
- ⇄ **Apple iWork** — Paste and Match Style `⌥⇧⌘V` ; Copy / Paste Style `⌥⌘C` / `⌥⌘V`

PLATFORM DIFFERENCES

- ▣ **Windows** — `F4` repeat, `F12` Save As
- ▣ **Mac** — `⌘Y` repeat; F-keys may need `fn`
- ▣ **Web** — `Ctrl+Shift+V` paste as text; no `F12`

PASTE OPTIONS

OPTION	KEEPS
`Keep Source Formatting`	Original fonts and colours
`Merge / Use Destination`	Text, takes styles of where you paste
`Keep Text Only`	Plain text; drops links and images
`Picture`	A static image of the selection
`Values (Excel)`	Results, not formulas

UNDO, REPEAT & SELECTION

ACTION	MAC / WINDOWS
`Undo history list`	Arrow next to Undo button
`Select all`	⌘A / Ctrl+A
`Extend selection`	⇧ + arrow keys
`Word / item jump`	⌥← → / Ctrl+← →
`Start / end of line`	⌘← → / Home End
`Start / end of doc`	⌘↑ ↓ / Ctrl+Home End

⚠ COMMON MISTAKES

- Pasting from a web page and fighting its fonts; paste as text from the start
- Clicking Format Painter once and expecting it to stay on; double-click it
- Re-doing a multi-step format by hand when `F4` / `⌘Y` repeats it

✓ TIPS & BEST PRACTICES

- Set the default paste (Word › Options › Advanced › Pasting from other programs) to Keep Text Only
- Use `F4` after inserting a row in Excel to insert more rows one keystroke at a time
- Turn on standard function keys on a Mac keyboard if you use Office shortcuts a lot

QUICK SUMMARY

- `^⌘V` / `Ctrl+Alt+V` — Paste Special
- `⌘Y` / `F4` — repeat last action
- `⇧⌘C / V` — copy / paste formatting
- `⇧F3` — cycle case

Sheet 06

Word Navigation & Selection

Navigation pane · Go To F5 · extend selection F8 · word / paragraph jumps · select sentence · Read Mode · Focus

MOVE THROUGH A LONG DOCUMENT

The **Navigation pane** (⌘F / Ctrl+F) lists every heading, so you can jump and even drag whole sections to reorder them. **Go To** (F5) jumps to a page, section, bookmark or comment. Selection shortcuts let you grab a word, sentence or paragraph without dragging, and **Focus** mode hides everything but the text.

NOTES

- Navigation pane has three views: **Headings**, **Pages** (thumbnails), **Results** (search hits)
- Drag a heading in the pane to move the heading and everything under it
- ⇧F5 / Shift+F5 returns to your last edit spot, even after reopening the file
- Click in the left margin: once = line, twice = paragraph, three times = whole document
- View › Focus (or Immersive Reader) for distraction-free writing

MOVING & SELECTING

```
⌘F              / Ctrl+F           # Navigation pane + search
⌥⌘G / F5        / F5 or Ctrl+G     # Go To (page, section, bookmark)
⇧F5             / Shift+F5         # back to last edit
⌥← →            / Ctrl+← →         # word by word
⌘↑ ↓            / Ctrl+↑ ↓         # paragraph by paragraph
⌘Home / End     / Ctrl+Home End    # start / end of document (fn⌘← → on laptops)
double-click                       # select word
⌘ + click       / Ctrl + click     # select sentence
triple-click                       # select paragraph
F8              / F8               # extend mode: F8 again = word, sentence…
⌥ + drag        / Alt + drag       # select a vertical block
```

CODE EXPLAINED

- → Extend mode: press F8 repeatedly to grow word → sentence → paragraph → document; Esc ends it
- → Go To accepts +3 or -2 to move relative pages
- → Holding ⇧ with any movement key turns it into a selection
- → ⌥ + drag block selection is handy for removing leading spaces or columns of text

OTHER SUITES

- ⇄ **Google Workspace** — Document outline (left pane); ^⌘N H / Ctrl+Alt+N H jumps to the next heading
- ⇄ **Apple iWork** — Pages: View › Table of Contents sidebar; ⌘ + click word selection the same

PLATFORM DIFFERENCES

- ▣ **Windows** — Ctrl+G / F5 Go To; Read Mode under View
- ▣ **Mac** — ⌥⌘G or **Edit › Find › Go To**; ⌘G is Find Next
- ▣ **Web** — Navigation pane under View; no F8 extend mode

GO TO TARGETS

TARGET	EXAMPLE
Page	12, +3, −1
Section	s2
Line	l40 (line 40)
Bookmark	name from Insert › Bookmark
Comment / footnote	next / previous
Table / graphic	cycle through them

VIEWS

VIEW	USE FOR
Print Layout	Normal editing with pages
Read Mode	Reading on screen, columns
Web Layout	No page breaks, wide tables
Outline	Restructuring by heading level
Draft	Fast typing, see section breaks
Focus	Full screen, text only

⚠ COMMON MISTAKES

- Scrolling through 80 pages to find a section instead of opening the Navigation pane
- Cut-and-pasting sections to reorder them; drag headings in the pane instead
- Using direct formatting for headings, so they never show up in the Navigation pane (use styles, sheet 07)

✓ TIPS & BEST PRACTICES

- Keep the Navigation pane open while writing long docs; it doubles as a live outline
- Use Outline view (⌥⌘O / Ctrl+Alt+O) with Show Level 2 to restructure fast
- Press ⇧F5 right after opening a document to jump to where you left off

QUICK SUMMARY

- ⌘F / Ctrl+F — Navigation pane
- F5 — Go To page / section / bookmark
- F8 — extend selection
- ⇧F5 — back to last edit

Sheet 07

Styles & Formatting

Styles gallery · Heading 1–3 · Clear Formatting · Update to Match · Style Inspector · Reveal Formatting · style sets

FORMAT WITH STYLES, NOT BY HAND

A **style** is a named bundle of formatting (font, size, spacing, numbering). Apply **Heading 1–3** and **Normal** instead of bolding and resizing text by hand, and you get a Navigation pane, an automatic table of contents, and the ability to restyle the whole document by changing one style. **Update to Match Selection** copies hand formatting back into the style.

NOTES

- Styles pane: Home › Styles launcher, or `⌥⌘⇧S` (Mac) / `Ctrl+Alt+Shift+S` (Win)
- Right-click a style › **Modify** to change it everywhere it's used
- **Update Heading 1 to Match Selection** turns one hand-formatted heading into the rule
- **Reveal Formatting** (`⇧F1`) shows exactly what's applied to the selected text
- Paragraph styles apply to whole paragraphs; character styles to selected words only

STYLE SHORTCUTS

```
⌥⌘1 2 3      / Ctrl+Alt+1 2 3  # Heading 1, 2, 3
⇧⌘N          / Ctrl+Shift+N    # Normal style
⌘⇧L          / Ctrl+Shift+L    # List Bullet style
^Space       / Ctrl+Space      # clear character formatting
⌘⌥⇧S         / Ctrl+Alt+Shift+S # Styles pane
⇧⌘S          / Ctrl+Shift+S    # Apply Styles box (type a name)
⇧F1          / Shift+F1        # Reveal Formatting
⌘[  ⌘]       / Ctrl+[  Ctrl+]  # shrink / grow font 1 pt
⇧⌘D          / Ctrl+Shift+D    # double underline
⇧⌘K          / Ctrl+Shift+K    # small caps
```

CODE EXPLAINED

→ Heading shortcuts make the text a heading **and** add it to the Navigation pane / TOC

→ `Ctrl+Space` removes character formatting; Home › Clear All Formatting (Aa eraser) removes both

→ The Apply Styles box (`Ctrl+Shift+S`) lets you type any style name, including custom ones

→ On Mac Word, `⇧⌘S` is Save As in some setups; check Tools › Customize Keyboard

OTHER SUITES

⇄ **Google Workspace** — `⌘⌥1–6` headings, Format › Paragraph styles › Update to match

⇄ **Apple iWork** — Pages paragraph styles in the Format sidebar; Update button when a style is changed

PLATFORM DIFFERENCES

- **Windows** — full Styles pane, Style Inspector, Manage Styles
- **Mac** — Styles pane; no Style Inspector
- **Web** — gallery + Modify; fewer options, no Reveal Formatting

BUILT-IN STYLES WORTH USING

STYLE	FOR
Normal	Body text; base for most styles
Heading 1–3	Structure, TOC, Navigation pane
Title / Subtitle	Cover page text (not in TOC)
List Bullet / List Number	Lists with consistent indents
Quote / Intense Quote	Pull quotes
Caption	Figure / table captions (sheet 08)

MODIFY STYLE OPTIONS

OPTION	EFFECT
Style based on	Inherits from another style
Style for following paragraph	Heading → Normal after Enter
Add to Styles gallery	Show on the Home tab
Automatically update	Hand edits change the style (risky)
New documents based on this template	Save to Normal.dotm

⚠ COMMON MISTAKES

- Making headings by bolding Normal text; they don't appear in the TOC or Navigation pane
- Turning on **Automatically update**; one stray edit restyles every matching paragraph
- Pressing Enter twice for spacing instead of setting Space After in the style

✓ TIPS & BEST PRACTICES

- Set paragraph spacing (Before / After) in styles and never use blank lines for space
- Use Style sets (Design tab) to try a new look without touching each style
- Save finished styles to a template (sheet 04) so every new doc starts right

QUICK SUMMARY

- `⌥⌘1` / `Ctrl+Alt+1` — Heading 1
- `⇧⌘N` / `Ctrl+Shift+N` — Normal
- `Update to Match` — copy hand formatting into a style
- `⇧F1` — Reveal Formatting

Sheet 08

Long Documents

heading-based TOC · captions · cross-references · section breaks · different first page · page numbering · footnotes · F9

REPORTS, THESES AND MANUALS

Long documents depend on **fields**: a **table of contents** built from heading styles, **captions** that number figures automatically, and **cross-references** that say "see Figure 3" and stay correct when figures move. **Section breaks** let one document have different margins, orientation, headers or page numbering in different parts.

NOTES

- TOC: References › Table of Contents; it lists Heading 1–3 by default
- Update all fields: select all (`⌘A` / `Ctrl+A`) then `F9`
- Section break (Next Page) vs page break: only a section break can change headers or orientation
- **Link to Previous** in a header ties it to the previous section; turn it off to make it different
- Footnote: `⌥⌘F` (Mac) / `Ctrl+Alt+F` (Win); endnote `⌥⌘E` / `Ctrl+Alt+D`

AUTOMATE IT

- Record a macro: select all, `F9`, then update TOC; run it before every export
- A template (sheet 04) with the cover, TOC and section setup already in place

LONG-DOCUMENT KEYS & FIELDS

```
Insert › Break / Ctrl+Enter   # page break
⌥⌘F          / Ctrl+Alt+F       # insert footnote
⌥⌘E          / Ctrl+Alt+D       # insert endnote
⌘A then F9   / Ctrl+A then F9   # update every field
⌥F9          / Alt+F9           # show / hide field codes

{ TOC \o "1-3" \h \z \u }      # table of contents (levels 1-3, links)
{ SEQ Figure \* ARABIC }        # caption number
{ REF _Ref123 \h }              # cross-reference to a bookmark
{ PAGE }  of  { NUMPAGES }      # "Page 3 of 20"
```

CODE EXPLAINED

- → Field codes in `{ }` are what Word stores; you insert them through menus, not by typing braces
- → `\o "1-3"` = heading levels in the TOC; `\h` makes entries clickable links
- → Captions (References › Insert Caption) create `SEQ` fields that renumber automatically
- → Cross-references (References › Cross-reference) point at captions, headings or bookmarks

OTHER SUITES

- ⇄ **Google Workspace** — Insert › Table of contents; Docs tabs for very long docs; no captions / cross-refs
- ⇄ **Apple iWork** — Pages: Insert › Table of Contents, section breaks, footnotes; no caption numbering fields

PLATFORM DIFFERENCES

- ▣ **Windows** — every feature, Navigation pane drag-to-reorder
- ▣ **Mac** — same features; some field shortcuts differ
- ▣ **Web** — TOC, footnotes, section breaks; limited header / numbering control

BREAKS

BREAK	DOES
`Page break`	Starts a new page, same section
`Section — Next Page`	New page + new section
`Section — Continuous`	New section on the same page (columns)
`Section — Odd / Even Page`	Chapters that start on right-hand pages
`Column break`	Jump to next column

HEADER / FOOTER OPTIONS

OPTION	USE FOR
`Different First Page`	No header on the cover
`Different Odd & Even`	Mirrored book layouts
`Link to Previous (off)`	Different header per chapter
`Format Page Numbers › Start at`	Restart at 1 after front matter
`Number format i, ii, iii`	Roman numerals for preface pages

⚠ COMMON MISTAKES

- Typing figure numbers by hand; insert captions so they renumber themselves
- Changing a header and having it change in every section; turn off **Link to Previous** first
- Exporting a PDF with a stale TOC; update fields (`F9`) right before you export

✓ TIPS & BEST PRACTICES

- Show formatting marks (`⌘8` / `Ctrl+Shift+8`) to see and delete stray section breaks
- Use **Update Table › Update entire table** after adding headings
- Put front matter in its own section with roman numerals and restart body numbering at 1

QUICK SUMMARY

- `TOC` — built from Heading styles
- `Captions + Cross-references` — auto numbering
- `Section break` — different headers / orientation
- `F9` — update fields

Sheet 09

Tables, Lists & Layout

Convert Text to Table · table styles · repeat header row · multilevel lists · columns · tab stops · margins per section

LINE THINGS UP PROPERLY

Use **tables** for anything in columns, **tab stops** for single lines with aligned parts (like a price list), and **multilevel lists** linked to headings for numbered sections like 1, 1.1, 1.1.1. Never line things up with spaces: they break the moment the font changes.

NOTES

- **Convert Text to Table** turns tab- or comma-separated text into a table instantly
- **Repeat Header Rows** (Table Layout) repeats the header on every page of a long table
- Tab in the last cell adds a row; `⇧Tab` goes back a cell
- Multilevel list: Home › Multilevel List › **Define New Multilevel List** › link levels to Heading 1–3
- Layout › Columns for newsletter columns; use a Continuous section break to mix 1 and 2 columns

TABLE & LIST KEYS

```
Tab / ⇧Tab          / Tab / Shift+Tab    # next / previous cell
Tab in last cell                          # add a row
⌥⇧↑ ↓               / Alt+Shift+↑ ↓      # move row / paragraph up / down
Layout › Split      / Ctrl+Shift+Enter    # split table here
^Tab                / Ctrl+Tab            # real tab inside a cell
⇧⌘L                 / Ctrl+Shift+L        # bulleted list
Tab / ⇧Tab (list)                         # demote / promote list level
1. + Space                                 # AutoFormat starts a numbered list
* + Space                                  # AutoFormat starts a bullet list
--- + Enter                               # horizontal line
```

CODE EXPLAINED

- → `Alt+Shift+↑` moves table rows and whole list items, not just text
- → Inside a table, `Tab` moves cells, so use `Ctrl+Tab` / `^Tab` to insert a real tab
- → AutoFormat As You Type creates lists and lines; undo once (`⌘Z`) to keep the plain text
- → Multilevel numbering stays correct only when every level uses the list's linked style

OTHER SUITES

- ⇄ **Google Workspace** — Format › Table; repeat header via Table properties › Pin header row
- ⇄ **Apple iWork** — Pages tables; header rows repeat on each page automatically

PLATFORM DIFFERENCES

- ▣ **Windows** — full table tools, Define New Multilevel List
- ▣ **Mac** — same; Table Design / Layout tabs
- ▣ **Web** — basic tables and lists; no list definitions or tab-stop dialog

WHICH TOOL?

NEED	USE
Rows and columns of data	Table
Name …… Price on one line	Right tab stop with dot leader
1 / 1.1 / 1.1.1 headings	Multilevel list linked to headings
Newsletter columns	Layout › Columns + section breaks
Side-by-side text blocks	Borderless table

TAB STOPS

TYPE	ALIGNS
Left	Text starts at the stop
Center	Text centred on the stop
Right	Text ends at the stop (page numbers)
Decimal	Decimal points line up
Leader ……	Dots fill the gap (Tabs dialog)

⚠ COMMON MISTAKES

- Spacing text into columns with the space bar; it misaligns with proportional fonts
- Restarting list numbers by typing them; right-click › Restart at 1 / Continue Numbering
- Long tables without Repeat Header Rows, so page 2 has no column labels

✓ TIPS & BEST PRACTICES

- Click and drag on the ruler to set tab stops; double-click a stop to open the Tabs dialog
- Use table styles (Table Design) instead of shading cells one by one
- Set **AutoFit to Window** so tables resize with margins

QUICK SUMMARY

- `Convert Text to Table` — tabs → table
- `Repeat Header Rows` — header on every page
- `Alt+Shift+↑↓` — move rows / items
- `Multilevel list` — 1 / 1.1 / 1.1.1

Sheet 10

Find & Replace, Quick Parts & AutoText

Advanced Find · wildcards · ^p ^t ^& · replace formatting · Quick Parts · Building Blocks · AutoText · AutoCorrect

NEVER TYPE THE SAME THING TWICE

Advanced Find & Replace can match special characters (^p paragraph marks, ^t tabs), formatting, and with **Use wildcards** on, patterns similar to regular expressions. **Quick Parts / AutoText** store reusable blocks (a signature, a disclaimer, a table) and **AutoCorrect** expands short codes into long text as you type.

NOTES

- Replace dialog: `^H` (Mac Word) / `Ctrl+H`; click **More** for special characters and formatting
- Replace `^p^p` with `^p` to remove double blank lines
- Wildcards: `<` `>` word start / end, `[0-9]{3}`, `(…)` groups, `\1` in Replace
- Save a block: select it › Insert › Quick Parts › **Save Selection to Quick Part Gallery**
- AutoText: type the entry name then press `F3` to insert it

AUTOMATE IT

- AutoText / Quick Parts for repeated paragraphs and tables
- AutoCorrect entry `addr1` → full mailing address (plain or formatted)

FIND & REPLACE CODES

```
^p      # paragraph mark          ^t   # tab
^l      # manual line break       ^m   # manual page break
^w      # any whitespace          ^?   # any character
^#      # any digit               ^$   # any letter
^&      # (Replace) the found text

# With "Use wildcards" ticked
<(Mr)>            # whole word Mr
[0-9]{3}-[0-9]{4}   # 555-1234
(<*>) (<*>)       → \2 \1    # swap two words
[A-Z]{2,}         # runs of capitals (acronyms)
```

CODE EXPLAINED

- → Codes like `^p` change when wildcards are on: use `^13` for paragraph marks in wildcard mode
- → `^&` in the Replace box reinserts what was found, e.g. find `TODO` replace `**^&**` with Bold format
- → Format button in the dialog searches for or replaces styles, fonts and highlighting
- → Wildcard `*` matches any run of characters, so `<*>` is "a whole word"

OTHER SUITES

- ⇄ **Google Workspace** — Edit › Find and replace › Match using regular expressions (true regex)
- ⇄ **Apple iWork** — Pages Find & Replace with options; no regex; Text Replacements in macOS settings

PLATFORM DIFFERENCES

- ▣ **Windows** — full Advanced Find, wildcards, Quick Parts, AutoText
- ▣ **Mac** — same features; Quick Parts via Insert › AutoText
- ▣ **Web** — basic Find & Replace, no wildcards or Quick Parts

WILDCARD PATTERNS

PATTERN	MATCHES
`?`	Any one character
`*`	Any run of characters
`[abc]` / `[a-z]`	One of / range
`[!0-9]`	Not a digit
`{n,m}`	n to m of the previous
`< >`	Start / end of word
`( ) + \1`	Group and reuse

REUSE TOOLS COMPARED

TOOL	INSERT BY	BEST FOR
AutoCorrect	Typing the code	Short text, typos
AutoText	Name + F3	Paragraphs
Quick Parts	Insert › Quick Parts	Formatted blocks, tables
Building Blocks	Organizer	Covers, headers, watermarks

⚠ COMMON MISTAKES

- Running Replace All without testing Find Next first; check a few hits before replacing
- Forgetting that **Use wildcards** stays ticked; normal searches then fail on `(` or `*`
- Saving Quick Parts to Building Blocks.dotx without saving it when Word asks on exit

✓ TIPS & BEST PRACTICES

- Replace formatting only: leave both boxes empty, set Format › Font on each
- Pick AutoCorrect codes that are never real words (`;sig`, `zaddr`)
- Highlight all matches: Find › Reading Highlight › Highlight All

QUICK SUMMARY

- `^p ^t ^&` — special codes
- `Use wildcards` — pattern search
- `Quick Parts` — reusable blocks
- `F3` — insert AutoText by name

Review & Collaboration

EDIT OTHER PEOPLE'S DOCUMENTS SAFELY

Track Changes records every insertion, deletion and format change with the author's name so the owner can **accept** or **reject** each one. **Comments** hold questions without changing text. **Compare** builds a tracked-changes view of two versions when someone edited without tracking. **Restrict Editing** limits a file to comments or form filling.

NOTES

- Markup view: **Simple Markup** (clean, red bars) or **All Markup** (every change shown)
- Accept / reject all at once: Review › Accept › Accept All Changes
- Lock tracking: Track Changes › **Lock Tracking** with a password
- **Editor** (Review tab) checks spelling, grammar, clarity and conciseness
- **Read Aloud** (Review tab; `Ctrl+Alt+Space` on Windows) is a quick proofreading pass

REVIEW SHORTCUTS

```
⇧⌘E          / Ctrl+Shift+E      # Track Changes on / off
⌥⌘A          / Ctrl+Alt+M        # new comment
—            / Alt R A C         # accept and move to next (KeyTips)
—            / Alt R J           # reject (KeyTips)
F7           / F7                # spelling & grammar
Review tab   / Ctrl+Alt+Space    # Read Aloud
⇧⌘G          / Ctrl+Shift+G      # word count

Review › Compare › Compare…      # two versions → tracked changes
Review › Compare › Combine…      # merge several reviewers' copies
```

CODE EXPLAINED

- → Turn tracking on **before** you edit; changes made before that aren't recorded
- → Reply to comments and **Resolve** them instead of deleting, so the thread stays visible
- → Compare creates a third document; neither original is changed
- → File › Info › Inspect Document removes leftover comments and tracked changes before sending

OTHER SUITES

- ⇄ **Google Workspace** — Suggesting mode (`⌘⌥⇧X`), comments `⌘⌥M`, Tools › Compare documents
- ⇄ **Apple iWork** — Pages Edit › Track Changes, comments; no compare tool

PLATFORM DIFFERENCES

- ▣ **Windows** — Compare, Combine, Lock Tracking, Restrict Editing
- ▣ **Mac** — same; Restrict Editing under Review › Protect
- ▣ **Web** — tracking and comments; Compare is desktop only

MARKUP VIEWS

VIEW	SHOWS
`Simple Markup`	Final text + red bar where changed
`All Markup`	Every insertion / deletion in colour
`No Markup`	What it looks like if all accepted
`Original`	What it looked like before

PROTECT OPTIONS

RESTRICTION	ALLOWS
`Tracked changes`	Edits, always tracked
`Comments`	Comments only
`Filling in forms`	Content controls / form fields
`No changes (Read only)`	Reading, optional exceptions
`Mark as Final`	Warning bar only, not security

⚠ COMMON MISTAKES

- Sending a "clean" file that still contains tracked changes hidden by No Markup view
- Deleting comments instead of resolving; you lose the discussion history
- Treating Mark as Final as protection; anyone can click Edit Anyway

✓ TIPS & BEST PRACTICES

- Run **Inspect Document** before sending outside your organization
- Set your name in Options › General so changes aren't attributed to "Author"
- Use Compare when a reviewer forgot Track Changes instead of reading both side by side

QUICK SUMMARY

- `⇧⌘E` / `Ctrl+Shift+E` — Track Changes
- `Accept / Reject` — per change or all
- `Compare` — tracked diff of two files
- `Inspect Document` — strip hidden data

Mail Merge & Fields

ONE LETTER, HUNDREDS OF RECIPIENTS

Mail merge combines a Word document with a list (usually an Excel table) to produce one letter, label, envelope or email per row. You insert **merge fields** like «FirstName» where data goes, add **Rules** for conditional text, preview, then **Finish & Merge** to a new document, the printer, or email via Outlook.

NOTES

- Data source: an Excel sheet with one header row and no blank rows works best
- Mailings › Start Mail Merge › Letters / Labels / Envelopes / E-mail Messages
- **Preview Results** shows real data; arrows step through records
- Rules › **If…Then…Else** inserts different text per record (e.g. Mr / Ms)
- Format numbers with field switches: `{ MERGEFIELD Amount \# "$#,##0.00" }`

AUTOMATE IT

- Build the merge once, save the document; next month just re-point it at the new list
- Email merge sends personalised messages through Outlook (desktop) in one step

MERGE FIELDS & SWITCHES

```
Dear «FirstName»,                          # merge field from the header row

{ MERGEFIELD Amount \# "$#,##0.00" }    # $1,250.00
{ MERGEFIELD DueDate \@ "MMMM d, yyyy" }  # March 4, 2026
{ IF { MERGEFIELD Balance } > 0 "Please pay by Friday." "Thank you!" }
{ DATE \@ "M/d/yyyy" }                     # today's date
{ PAGE }                                   # current page number

⌥F9       / Alt+F9        # toggle field codes
⇧F9       / Shift+F9      # toggle the selected field only
⌘⇧F9      / Ctrl+Shift+F9  # unlink: turn field into plain text
⌘F9       / Ctrl+F9       # insert empty field braces
```

CODE EXPLAINED

- → Excel formatting doesn't always carry over; use `\#` (number) and `\@` (date) switches
- → Insert real field braces with `⌘F9` / `Ctrl+F9`; typed `{ }` characters don't work
- → IF fields compare values and pick text, so one template handles several cases
- → Unlinking (`Ctrl+Shift+F9`) freezes a field's current result as text

OTHER SUITES

- ⇄ **Google Workspace** — Docs mail merge (Workspace plans) or an Apps Script merge (sheet 27–28 of that set)
- ⇄ **Apple iWork** — Pages mail merge from Numbers or Contacts (Format › Mail Merge)

PLATFORM DIFFERENCES

- ▣ **Windows** — full wizard, email merge through Outlook
- ▣ **Mac** — merge works; email merge needs Outlook for Mac
- ▣ **Web** — no mail merge

MERGE STEPS

STEP	MAILINGS TAB BUTTON
1 Pick type	Start Mail Merge
2 Connect list	Select Recipients › Use an Existing List
3 Filter / sort	Edit Recipient List
4 Add fields	Insert Merge Field / Address Block
5 Check	Preview Results + Check for Errors
6 Output	Finish & Merge

PICTURE SWITCHES

SWITCH	RESULT
`\# "#,##0"`	12,500
`\# "0.0%"`	12.5%
`\@ "dddd"`	Tuesday
`\@ "MMM-yy"`	Mar-26
`\* Upper`	ALL CAPS
`\* FirstCap`	First letter capital

⚠ COMMON MISTAKES

- Blank rows or merged cells in the Excel list, which stop or skew the merge
- Trusting Excel's number formatting; zip codes lose leading zeros unless the column is text
- Merging straight to the printer without previewing a few records first

✓ TIPS & BEST PRACTICES

- Keep the list as an Excel **table** (sheet 16) so new rows are included automatically
- Merge to a new document first and skim it before printing or emailing
- Use **Check for Errors** to catch empty or mismatched fields

QUICK SUMMARY

- `Mailings` — the merge workflow
- `«Field»` — data from the header row
- `\# \@` — number / date formats
- `⌥F9` / `Alt+F9` — show field codes

Sheet 13

Excel Navigation & Selection

⌘↓ / Ctrl+↓ · ⇧ extend · Ctrl+Home / End · Name Box · Go To Special · Freeze Panes · select row / column

GET AROUND BIG SHEETS WITHOUT SCROLLING

`Ctrl+Arrow` (`⌘+Arrow` on Mac) jumps to the edge of the current block of data; add `Shift` to select as you jump. The **Name Box** (left of the formula bar) jumps to any address or named range. **Go To Special** selects only blanks, formulas, constants or visible cells, which makes bulk fixes fast. **Freeze Panes** keeps headers on screen.

NOTES

- `Ctrl+Home` → A1; `Ctrl+End` → last used cell (shows leftover junk if it lands too far out)
- Type `B2:F500` or a range name in the Name Box and press Enter to select it
- Go To Special: `F5` / `Ctrl+G` › Special… (Mac: `^G` › Special)
- Select visible cells only: `⌘⇧Z` (Mac) / `Alt+;` (Win), then copy filtered rows
- Freeze: View › Freeze Panes freezes rows above and columns left of the active cell

MOVING & SELECTING

```
⌘↑ ↓ ← →        / Ctrl+↑ ↓ ← →     # jump to data edge
⇧⌘↓             / Ctrl+Shift+↓     # select to data edge
fn^←            / Ctrl+Home        # go to A1
fn^→            / Ctrl+End         # last used cell
^Space          / Ctrl+Space       # select column
⇧Space          / Shift+Space      # select row
⌘A              / Ctrl+A           # current region, again = whole sheet
⌥→ ←            / Ctrl+PgDn PgUp   # next / previous sheet
^G              / F5 or Ctrl+G     # Go To (then Special…)
⌘⇧Z             / Alt+;            # select visible cells only
^⌘\ / Ctrl+\   / Ctrl+\           # row differences
```

CODE EXPLAINED

- → Jumps stop at the first blank cell, so gaps in your data break `Ctrl+↓`
- → `Ctrl+A` inside data selects the block (current region); press again for the whole sheet
- → After Go To Special › Blanks, type `=↑` and press `Ctrl+Enter` to fill each blank from the cell above
- → Mac laptop shortcuts often need `fn` for Home / End / Page keys

OTHER SUITES

- ⇄ **Google Workspace** — same `⌘↓` / `Ctrl+↓` jumps; no Go To Special (use filters)
- ⇄ **Apple iWork** — Numbers: tables are separate objects; `⌥↑↓` add rows, `⌘↑↓` table edge

PLATFORM DIFFERENCES

- ▣ **Windows** — every shortcut, Go To Special
- ▣ **Mac** — same ideas, `fn` for Home / End, different sheet-switch keys
- ▣ **Web** — arrows and Name Box work; Go To Special limited

GO TO SPECIAL PICKS

OPTION	SELECTS
Blanks	Empty cells in the selection
Constants	Typed values (not formulas)
Formulas	Cells with formulas, by result type
Visible cells only	Skips hidden / filtered rows
Last cell	Same as Ctrl+End
Conditional formats / Data validation	Cells with rules

FREEZE PANES

WANT	CLICK IN	THEN
Header row	any cell	Freeze Top Row
First column	any cell	Freeze First Column
Rows 1–2 + column A	B3	Freeze Panes
Undo	—	Unfreeze Panes

⚠ COMMON MISTAKES

- Leaving stray formatting far below the data, so `Ctrl+End` and the scroll bar go to row 50,000
- Copying a filtered range on the web and pasting hidden rows too; select visible cells first
- Clicking A2 and choosing Freeze Panes when you wanted column A too; click B2

✓ TIPS & BEST PRACTICES

- Delete unused rows / columns below and right of the data, then save, to reset `Ctrl+End`
- Name important ranges (sheet 15) and jump to them from the Name Box
- Use `Ctrl+Shift+↓` then `Ctrl+Shift+→` to select a whole table in two keystrokes

QUICK SUMMARY

- `⌘↓` / `Ctrl+↓` — jump to edge
- `⇧` + jump — select
- `F5 › Special` — blanks, formulas, visible
- `Freeze Panes` — keep headers visible

Sheet 14

Data Entry, Fill & Validation

Flash Fill Ctrl+E · Fill Down ⌘D / Ctrl+D · fill series · Ctrl+Enter · Ctrl+; · Data Validation · checkboxes

TYPE LESS, FILL MORE

Excel fills a lot for you. **Flash Fill** watches one or two examples and fills the rest of the column (split names, reformat IDs). **Fill Down** copies the cell above; the **fill handle** extends series like dates and weekdays. `Ctrl+Enter` puts the same entry into every selected cell. **Data Validation** limits what can be typed and can add a dropdown list.

NOTES

- Flash Fill: type the first result, then `Ctrl+E` (or Data › Flash Fill)
- Drag the fill handle with the right mouse button for options (fill weekdays, months, series)
- Double-click the fill handle to fill down as far as the column next to it has data
- Insert › **Checkbox** puts real TRUE / FALSE checkboxes in cells (Microsoft 365)
- Custom lists for fill: Options › Advanced › Edit Custom Lists

AUTOMATE IT

- Flash Fill handles one-off cleanups; record a macro or Office Script (sheet 35) when it repeats every week
- Dropdowns from a table column grow automatically when you add list items

ENTRY SHORTCUTS

```
⌘D  ⌘R          / Ctrl+D  Ctrl+R     # fill down / fill right
Data › Flash Fill / Ctrl+E           # Flash Fill
^↩              / Ctrl+Enter         # enter in every selected cell
^;              / Ctrl+;             # today's date
⌘;              / Ctrl+Shift+;       # current time
⌥↓              / Alt+↓              # pick from values above / dropdown
⌥⌘↩ / ^⌥↩       / Alt+Enter          # new line inside a cell
F2 / ^U         / F2                 # edit cell
⌘Z after entry                       # undo an AutoCorrect / AutoFill

Data › Data Validation › Allow: List › Source: =Status   # dropdown from named range
```

CODE EXPLAINED

- → `Ctrl+Enter` with a range selected fills all of it, formulas included, with relative refs adjusted
- → `Alt+↓` in an empty cell lists values already typed in that column
- → Point a validation list at a table column (via a name) so the dropdown updates itself
- → Checkboxes store TRUE / FALSE, so `=COUNTIF(B2:B50,TRUE)` counts ticked boxes

OTHER SUITES

- ⇄ **Google Workspace** — Smart Fill `⌘⇧Y`, dropdown chips, Insert › Checkbox
- ⇄ **Apple iWork** — Numbers autofill handle, pop-up menus and checkboxes as cell formats

PLATFORM DIFFERENCES

- ▣ **Windows** — everything; Flash Fill `Ctrl+E`
- ▣ **Mac** — same; some entry keys differ (`⌘;` time)
- ▣ **Web** — Flash Fill, validation and checkboxes; fewer validation options

FILL HANDLE RESULTS

START WITH	DRAG GIVES
`Mon`	Tue, Wed, Thu…
`Jan / January`	Feb, Mar…
`1, 2 (both selected)`	3, 4, 5…
`Q1`	Q2, Q3, Q4, Q1…
`Item 1`	Item 2, Item 3…
`1/31/2026`	next days (right-drag › Fill Months)

VALIDATION TYPES

ALLOW	EXAMPLE
`List`	Open, Closed, On hold
`Whole number / Decimal`	between 1 and 100
`Date`	after `=TODAY()`
`Text length`	≤ 10 characters
`Custom`	`=COUNTIF($A:$A,A2)=1` (no duplicates)

⚠ COMMON MISTAKES

- Flash Fill results are static; they don't update when the source changes (use formulas for live data)
- Typing dropdown items into the validation Source box, then having to edit every copy later
- Gaps in the neighbouring column stop double-click fill early

✓ TIPS & BEST PRACTICES

- Add an **Input Message** and **Error Alert** on the validation tabs so users know the rule
- Use Circle Invalid Data (Data Validation menu) to find bad values already in the sheet
- Format the column before entering IDs with leading zeros (Text format)

QUICK SUMMARY

- `Ctrl+E` — Flash Fill
- `⌘D` / `Ctrl+D` — fill down
- `Ctrl+Enter` — fill selection
- `Data Validation › List` — dropdown

Sheet 15

Formula Basics

operators · $ absolute refs · ⌘T / F4 · cross-sheet refs · named ranges · #N/A #VALUE! #SPILL! #REF! · Evaluate Formula

HOW FORMULAS THINK

Every formula starts with =. References are **relative** by default (they shift when copied); add $ to lock the column, row, or both. **Named ranges** (`TaxRate` instead of $B$1) make formulas readable. When something breaks, the **error value** tells you why, and **Evaluate Formula** and **Trace Precedents** show how Excel got there.

NOTES

- Toggle `$` while editing a reference: `F4` (Win) or `⌘T` (Mac) cycles `A1 → $A$1 → A$1 → $A1`
- Other sheet: `=Sales!B2`; other sheet with spaces: `='Q1 Sales'!B2`
- Name a cell: click it, type a name in the Name Box, Enter. Manage in Formulas › Name Manager
- Show formulas instead of results: **^ + grave accent** (Mac) / **Ctrl + grave accent** (Win)
- Formulas › Evaluate Formula steps through one calculation (Windows)

REFERENCES & OPERATORS

```
=B2*C2                  # relative: copied down becomes =B3*C3
=B2*$F$1                # $F$1 stays fixed (tax rate)
=B$1*$A2                # mixed: row 1 fixed, column A fixed (times table)
=SUM(Sales!B2:B13)      # another sheet
=SUM('Q1 Sales'!B:B)    # sheet name with spaces needs quotes
=Price*(1+TaxRate)      # named ranges
="Total: "&TEXT(D2,"$#,##0")    # & joins text → Total: $1,250
=A2<>""                 # TRUE if A2 not empty
=(B2-C2)/C2             # parentheses before / → growth %
=2^10                   # 1024
```

CODE EXPLAINED

- → Order: `( )`, `^`, `* /`, `+ -`, `&`, then comparisons `= < > <= >= <>`
- → `$` only matters when copying; it doesn't change the result in place
- → Names are workbook-wide by default, so `TaxRate` works on every sheet
- → `&` joins text; numbers turn into text with no formatting unless you wrap them in `TEXT()`

OTHER SUITES

- ⇄ **Google Workspace** — same syntax and `$`; `F4` toggles refs; sheet refs `'Sheet 1'!A1`
- ⇄ **Apple iWork** — Numbers uses header names and `Table 1::B2`; preserve row / column instead of `$`

PLATFORM DIFFERENCES

- ▣ **Windows** — `F4`, Evaluate Formula, Watch Window
- ▣ **Mac** — `⌘T` for `$`; Evaluate Formula missing in some builds
- ▣ **Web** — `F4` works; tracing arrows limited

ERROR VALUES

ERROR	USUALLY MEANS
`#N/A`	Lookup found no match
`#VALUE!`	Wrong type: text where a number was expected
`#REF!`	A referenced cell was deleted
`#DIV/0!`	Dividing by zero or a blank
`#NAME?`	Typo in a function or range name
`#SPILL!`	Dynamic array result is blocked by data
`#NUM!`	Impossible number (e.g. SQRT(-1))

FORMULA AUDITING

TOOL	SHOWS
`Trace Precedents`	Arrows from cells this formula uses
`Trace Dependents`	Arrows to formulas using this cell
`Evaluate Formula`	Step-by-step calculation
`Watch Window`	Live values from far-away cells
`Error Checking`	Next error and its cause
`^[ / Ctrl+[`	Jump to precedents

⚠ COMMON MISTAKES

- Copying a formula and forgetting `$`, so the tax-rate reference walks down the sheet
- Numbers stored as text (green triangle); they're ignored by SUM. Convert via the warning menu
- Deleting a row that formulas point to and getting #REF!; check dependents first

✓ TIPS & BEST PRACTICES

- Click a reference in the formula bar and press `F4` / `⌘T` until the `$` is where you want it
- Name inputs (rates, thresholds) so formulas explain themselves
- Use `Ctrl+[` to jump to a formula's source cell; `F5` + Enter jumps back

QUICK SUMMARY

- `$A$1` — absolute
- `F4` / `⌘T` — cycle `$`
- `Sheet!A1` — cross-sheet
- `#N/A #REF! #SPILL!` — read the error

Sheet 16

Tables & Structured References

Format as Table ^T / Ctrl+T · Table1[Column] · [@Column] · Total Row · auto-expanding · slicers · table vs range

RANGES THAT GROW BY THEMSELVES

An Excel **table** (not just a formatted range) knows its own size. New rows are added automatically, formulas in a column fill down by themselves, and you refer to data by name: `Sales[Amount]` instead of `$C$2:$C$500`. PivotTables, charts, dropdowns and Power Query that point at a table pick up new rows without edits.

NOTES

- Create: click in the data › `^T` (Mac) / `Ctrl+T` (Win) › tick **My table has headers**
- Rename it in Table Design › Table Name (e.g. `Sales`) before writing formulas
- `[@Amount]` = the Amount value on this row; `Sales[Amount]` = the whole column
- **Total Row** (Table Design) adds a SUBTOTAL row with a dropdown per column
- **Slicers** (Table Design › Insert Slicer) are clickable filter buttons

AUTOMATE IT

- Point PivotTables, charts and validation lists at tables so new rows are picked up automatically
- A table-based monthly import + Refresh All beats re-selecting ranges every month

STRUCTURED REFERENCES

```
=[@Qty]*[@Price]                  # row formula; fills the whole column
=SUM(Sales[Amount])               # whole column, grows with the table
=SUMIFS(Sales[Amount], Sales[Region], "West")
=COUNTA(Sales[OrderID])           # number of rows
=Sales[[#Totals],[Amount]]        # value in the Total Row
=Sales[#Headers]                  # header row
=XLOOKUP(A2, Products[SKU], Products[Price])
=ROWS(Sales)                      # rows in the data body
=Sales[@[Qty]:[Price]]            # Qty through Price on this row
```

CODE EXPLAINED

- → Excel writes these for you: click the cells while typing a formula inside or outside the table
- → Structured references don't need `$`; they don't shift when copied across columns of other tables
- → A calculated column (formula in a table column) is one formula, kept consistent for every row
- → The Total Row uses `SUBTOTAL(109, …)`, so it ignores filtered-out rows

OTHER SUITES

- ⇄ **Google Workspace** — Format › Convert to table `⌘⌥T`; `Table1[Column]` references
- ⇄ **Apple iWork** — every Numbers table already has header rows; formulas use header names

PLATFORM DIFFERENCES

- ▣ **Windows** — full tables + slicers
- ▣ **Mac** — same; table shortcut is `^T`
- ▣ **Web** — create and use tables, slicers on newer builds

REFERENCE PARTS

SYNTAX	MEANS
`Table[Col]`	Data in one column
`[@Col]`	This row, one column
`Table[#All]`	Headers + data + totals
`Table[#Data]`	Data body only
`Table[#Totals]`	Total row
`Table[[ColA]:[ColC]]`	Range of columns

TABLE VS PLAIN RANGE

FEATURE	TABLE	RANGE
New rows included	Yes	No
Formulas auto-fill	Yes	No
Readable refs	Yes	No
Banded rows / filter	Built in	Manual
Merged cells allowed	No	Yes

⚠ COMMON MISTAKES

- Leaving the default name `Table1`; rename tables before formulas reference them
- Blank rows in the middle of data; the table still works but your eyes and filters suffer
- Typing values over a calculated column, which breaks the column's consistency

✓ TIPS & BEST PRACTICES

- Press `Tab` in the last cell to add a row
- Use **Convert to Range** if a file must go to an app that can't read tables
- Turn off auto-fill of calculated columns per formula with the lightning-bolt button

QUICK SUMMARY

- `Ctrl+T` — make a table
- `Table[Col]` — whole column
- `[@Col]` — this row
- `Total Row` — SUBTOTAL that respects filters

Logic & Math Functions

DECIDE AND ADD UP

IF returns one value when a test is TRUE and another when FALSE. IFS and SWITCH replace deep nested IFs. The **…IFS** family (**SUMIFS, COUNTIFS, AVERAGEIFS, MAXIFS, MINIFS**) adds or counts only rows that match one or more conditions, which covers most reporting. SUMPRODUCT multiplies arrays and adds the result.

NOTES

- Criteria are text: `">100"`, `"<>Closed"`, `">="&F1` (join a cell)
- Wildcards in criteria: `"*north*"`, `"A?"`; escape a literal `*` with `~*`
- SUMIFS argument order: **sum range first**, then range / criteria pairs
- `IFERROR(x, "")` hides every error; `IFNA` only #N/A (safer for lookups)
- Round results you compare: `ROUND(x, 2)` avoids floating-point surprises

LOGIC & CONDITIONAL MATH

```
=IF(C2>=70, "Pass", "Fail")                          # → Pass
=IFS(C2>=90,"A", C2>=80,"B", C2>=70,"C", TRUE,"F")   # TRUE = else
=SWITCH(B2, "N","North", "S","South", "Other")       # last arg = default
=IF(AND(B2="West", C2>1000), "Big West", "")
=IF(OR(D2="", E2=""), "Missing", "OK")
=SUMIFS(Sales[Amount], Sales[Region],"West", Sales[Date],">="&F1)
=COUNTIFS(A:A, "Open", B:B, "<"&TODAY())             # overdue open items
=AVERAGEIFS(C:C, B:B, "Widget")
=MAXIFS(C:C, B:B, "Widget")                          # also MINIFS
=SUMPRODUCT(Qty, Price)                              # Σ qty × price
=ROUND(B2*1.0825, 2)    =ROUNDUP(x,0)    =MROUND(x, 0.25)
=IFERROR(B2/C2, 0)                                   # 0 instead of #DIV/0!
```

CODE EXPLAINED

- → IFS has no default; end with `TRUE, value` to catch everything else
- → Comparison operators in criteria must be inside the quotes; cell values are joined with `&`
- → `COUNTIFS` with `"<"&TODAY()` compares against today's date number
- → `SUMPRODUCT` also handles conditions: `=SUMPRODUCT((Region="West")*Amount)`

OTHER SUITES

- ⇄ **Google Workspace** — same functions and argument order; also `SUMIF` / `COUNTIF` with regex via `REGEXMATCH`
- ⇄ **Apple iWork** — Numbers has IF, IFS, SWITCH, SUMIFS, COUNTIFS, AVERAGEIFS, IFERROR

PLATFORM DIFFERENCES

- ▣ Same on all three. Formulas are the most portable part of Excel.

CRITERIA EXAMPLES

CRITERIA	MATCHES
`"West"`	Exactly West (not case-sensitive)
`">100"`	Greater than 100
`"<>"`	Not blank
`""`	Blank
`"*north*"`	Contains north
`">="&F1`	At least the value in F1

THE …IFS FAMILY

FUNCTION	RETURNS
`SUMIFS`	Sum of matching rows
`COUNTIFS`	Count of matching rows
`AVERAGEIFS`	Average of matching rows
`MAXIFS / MINIFS`	Largest / smallest match
`COUNTIF(range, x)>1`	Duplicate test

⚠ COMMON MISTAKES

- Putting the sum range last in SUMIFS (that's the old SUMIF order)
- Writing `">F1"`; that compares to the text "F1". Use `">"&F1`
- Wrapping everything in IFERROR, hiding real mistakes like typos in ranges

✓ TIPS & BEST PRACTICES

- Use SUMIFS even with one condition; same order every time and easy to add more
- Build criteria from cells (`">="&StartDate`) so reports are driven by inputs, not edits
- Prefer IFS or SWITCH over IFs nested more than two deep

QUICK SUMMARY

- `IF / IFS / SWITCH` — choose a value
- `SUMIFS` — sum range first, then pairs
- `">"&cell` — criteria from a cell
- `IFNA` — catch only lookup misses

Lookup Functions

XLOOKUP · XMATCH · INDEX / MATCH · VLOOKUP (legacy) · approximate & wildcard match · two-way lookup · CHOOSECOLS

FIND A VALUE IN ANOTHER TABLE

XLOOKUP is the modern lookup: give it the value, the column to search and the column to return. It looks left or right, defaults to an exact match, has a built-in "not found" value, and can return several columns at once. **INDEX / MATCH** does the same in older versions. **VLOOKUP** still works but breaks when columns are inserted.

NOTES

- `XLOOKUP(lookup, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`
- match_mode: `0` exact (default), `-1` exact or next smaller, `1` next larger, `2` wildcard
- search_mode: `-1` searches last to first (latest entry)
- Return several columns: make return_array span them; results spill right
- XLOOKUP and XMATCH are in Excel 2021 and later, Microsoft 365 and the web

LOOKUP PATTERNS

```
=XLOOKUP(A2, Products[SKU], Products[Price])                  # exact match
=XLOOKUP(A2, Products[SKU], Products[Price], "Not found")     # custom miss
=XLOOKUP(A2, Products[SKU], Products[[Name]:[Price]])         # spills 3 columns
=XLOOKUP(D2, Rates[MinIncome], Rates[Rate], , -1)             # tax bracket
=XLOOKUP(A2, Log[ID], Log[Status], , 0, -1)                   # last match
=XLOOKUP("*"&B2&"*", Names, Emails, , 2)                      # contains
=XLOOKUP(G2, Months, XLOOKUP(G1, Regions, Data))              # two-way
=INDEX(Products[Price], MATCH(A2, Products[SKU], 0))          # classic
=XMATCH(A2, Products[SKU])                                    # position → 7
=VLOOKUP(A2, $F$2:$H$200, 3, FALSE)                           # legacy
```

CODE EXPLAINED

- → Leave optional arguments empty with commas: `, , -1`
- → Approximate match (`-1`) needs no sorting in XLOOKUP, unlike VLOOKUP's TRUE mode
- → Nested XLOOKUP: the inner one returns a whole column, the outer one picks the row
- → VLOOKUP's column number `3` silently changes meaning when a column is inserted

OTHER SUITES

- ⇄ **Google Workspace** — XLOOKUP, XMATCH, INDEX/MATCH, VLOOKUP all supported; FILTER for multi-row results
- ⇄ **Apple iWork** — Numbers supports XLOOKUP, XMATCH, INDEX/MATCH, VLOOKUP, LOOKUP

PLATFORM DIFFERENCES

- ▣ **Windows / Mac / Web** — XLOOKUP in M365 and 2021+; Office 2019 and older need INDEX/MATCH

XLOOKUP VS VLOOKUP

	XLOOKUP	VLOOKUP
Default match	Exact	Approximate (!)
Look left	Yes	No
Survives inserted columns	Yes	No
Not-found value	Built in	Wrap in IFNA
Return many columns	Yes	One

MATCH MODES

MODE	FINDS
`0`	Exact (default)
`-1`	Exact or next smaller
`1`	Exact or next larger
`2`	Wildcards * ? ~
`search -1`	Last match first
`search 2 / -2`	Binary search (sorted data)

⚠ COMMON MISTAKES

- VLOOKUP without `FALSE`; it defaults to approximate and returns wrong rows on unsorted data
- Lookup values with trailing spaces or numbers stored as text; wrap with `TRIM()` or fix the type
- Return and lookup arrays of different sizes; XLOOKUP returns #VALUE!

✓ TIPS & BEST PRACTICES

- Use table columns as arrays so lookups never need `$` and grow with the data
- Use the not-found argument instead of wrapping in IFERROR, so real errors still show
- For many matching rows, use FILTER (sheet 19) instead of a lookup

QUICK SUMMARY

- `XLOOKUP(val, look, return)` — the default
- `, , -1` — next smaller (brackets)
- `INDEX/MATCH` — older versions
- `VLOOKUP(…, FALSE)` — legacy only

Dynamic Arrays

ONE FORMULA, MANY RESULTS

A **dynamic array** formula returns several values that **spill** into the cells below and to the right. `FILTER` pulls matching rows, `SORT` and `UNIQUE` reorder and dedupe, and `SEQUENCE` generates numbers. Refer to a whole spilled result with the **spill operator** `#`, e.g. `=COUNTA(E2#)`, so dependent formulas resize with it.

NOTES

- Only the top-left cell holds the formula; the rest show greyed "ghost" values
- `#SPILL!` means something is in the way; clear the cells shown by the dashed border
- Spilling doesn't work inside tables; put dynamic formulas next to the table
- `@` in old formulas forces a single value (implicit intersection)
- Microsoft 365 and Excel 2021+; TAKE / DROP / VSTACK / TOCOL need current M365 or 2024

SPILL FORMULAS

```
=FILTER(Sales, Sales[Region]="West")                  # all West rows
=FILTER(Sales, (Sales[Region]="West")*(Sales[Amount]>1000), "None")  # AND
=FILTER(Sales, (Sales[Region]="West")+(Sales[Region]="East"))        # OR
=SORT(Sales, 3, -1)                                   # by 3rd column, descending
=SORTBY(Sales[Name], Sales[Amount], -1)               # names by amount
=UNIQUE(Sales[Region])                                # distinct list
=SORT(UNIQUE(Sales[Region]))                          # sorted distinct
=COUNTA(UNIQUE(Sales[Customer]))                      # distinct count
=SEQUENCE(12)          =SEQUENCE(1, 7, 0)             # 1…12 down / 0…6 across
=TAKE(SORT(Sales,3,-1), 5)                            # top 5 rows
=VSTACK(Jan, Feb, Mar)                                # stack tables
=E2#                                                  # whole spill from E2
```

CODE EXPLAINED

- → Multiply conditions for AND, add them for OR; each comparison gives TRUE / FALSE (1 / 0)
- → The third FILTER argument is shown when nothing matches, instead of `#CALC!`
- → `TAKE(x, 5)` keeps the first 5 rows; `TAKE(x, -5)` the last 5; `DROP` removes them
- → Chart or validation sources can point at `=$E$2#` and resize automatically

OTHER SUITES

- ⇄ **Google Workspace** — FILTER, SORT, UNIQUE, SEQUENCE, VSTACK spill natively; older sheets use ARRAYFORMULA
- ⇄ **Apple iWork** — recent Numbers has FILTER, SORT, UNIQUE, SEQUENCE, TAKE, VSTACK too

PLATFORM DIFFERENCES

- ▣ **Windows / Mac / Web** — same in M365; Office 2019 and older have no spilling

ARRAY FUNCTIONS

FUNCTION	DOES
`FILTER(arr, include, [if_empty])`	Rows that match
`SORT(arr, [col], [order])`	Sort by column index
`SORTBY(arr, by, [order])`	Sort by another array
`UNIQUE(arr, [by_col], [once])`	Distinct / appear-once
`SEQUENCE(rows, [cols], [start], [step])`	Number grid
`TAKE / DROP`	Keep / remove first or last n
`VSTACK / HSTACK`	Combine arrays
`CHOOSECOLS / CHOOSEROWS`	Pick columns / rows
`TOCOL / TOROW`	Flatten to one column / row

OLD WAY → NEW WAY

OLD	NEW
`Advanced Filter / copy-paste`	FILTER
`Remove Duplicates (static)`	UNIQUE (live)
`Helper rank columns`	SORTBY + TAKE
`Ctrl+Shift+Enter arrays`	Just Enter
`OFFSET dynamic ranges`	E2# spill refs

⚠ COMMON MISTAKES

- Typing in a cell inside a spill range and getting #SPILL! on the formula
- Putting FILTER inside a table, where spilling isn't allowed
- Referencing `E2:E50` instead of `E2#`, so new results fall outside your range

✓ TIPS & BEST PRACTICES

- Use FILTER + SORT + TAKE for live "top 10" lists on dashboards
- Wrap UNIQUE in SORT for clean dropdown sources
- Select the formula cell and look for the blue spill border to see the full result

QUICK SUMMARY

- `FILTER` — matching rows
- `SORT / SORTBY / UNIQUE` — reorder, dedupe
- `E2#` — the whole spill
- `#SPILL!` — clear the blocked cells

Text, Date & Regex Functions

CLEAN TEXT AND DO DATE MATH

Modern text functions split and join without nested FIND / MID: TEXTSPLIT, TEXTBEFORE, TEXTAFTER, TEXTJOIN. The **REGEX** functions match patterns. Dates are numbers (days since 1900), so subtracting gives days, and EOMONTH, EDATE, NETWORKDAYS handle month ends, adding months and business days.

NOTES

- `TEXT(value, format)` turns numbers / dates into formatted text
- Dates are serial numbers: `=B2-A2` is days between; format the result as Number
- `TRIM` removes extra spaces; `CLEAN` removes non-printing characters
- REGEX functions are in Microsoft 365 (2024 onward), including the web
- `DATEDIF` is undocumented but works: `"y"`, `"m"`, `"d"`, `"ym"`

TEXT & DATE FORMULAS

```
=TEXTSPLIT("Smith, Jane", ", ")          # Smith | Jane (spills)
=TEXTBEFORE(A2, "@")                     # jane  (from jane@acme.com)
=TEXTAFTER(A2, "@")                      # acme.com
=TEXTJOIN(", ", TRUE, B2:B10)            # join, skipping blanks
=PROPER(TRIM(A2))                        # "  jANE  smith" → Jane Smith
=TEXT(B2, "$#,##0.00")   =TEXT(C2, "mmm d, yyyy")
=REGEXTEST(A2, "^\d{5}$")                # TRUE for 5-digit zip
=REGEXEXTRACT(A2, "\d{3}-\d{4}")         # 555-1234
=REGEXREPLACE(A2, "\s+", " ")            # collapse whitespace
=EOMONTH(TODAY(), 0)                     # last day of this month
=EDATE(A2, 3)                            # 3 months later
=NETWORKDAYS(A2, B2, Holidays)           # business days
=DATEDIF(Birth, TODAY(), "y")            # age in years
=WORKDAY(TODAY(), 10)                    # 10 business days out
```

CODE EXPLAINED

- → TEXTSPLIT spills across columns; a second delimiter argument splits into rows too
- → TEXTBEFORE / TEXTAFTER take an instance number: `TEXTAFTER(A2, "/", -1)` = after the last slash
- → In REGEX patterns `\d` digit, `\s` space, `^ $` start / end; the functions are case-sensitive by default
- → `EOMONTH(date, -1)+1` gives the first day of the current month

OTHER SUITES

- ⇄ **Google Workspace** — SPLIT, REGEXMATCH / REGEXEXTRACT / REGEXREPLACE, TEXTJOIN, EOMONTH, NETWORKDAYS
- ⇄ **Apple iWork** — Numbers: TEXTBEFORE / TEXTAFTER, REGEX inside functions, REGEX.EXTRACT, DATEDIF

PLATFORM DIFFERENCES

- ▣ **M365 (all)** — REGEX and TEXTSPLIT in current builds; Office 2021 lacks TEXTSPLIT, 2024 lacks REGEX

TEXT FORMAT CODES

CODE	RESULT
`0.0`	1234.5
`#,##0`	1,235
`0%`	45%
`mm/dd/yyyy`	03/04/2026
`dddd`	Wednesday
`mmm yyyy`	Mar 2026
`h:mm AM/PM`	2:30 PM

DATE FUNCTIONS

FUNCTION	RETURNS
`TODAY() / NOW()`	Date / date + time
`DATE(y, m, d)`	Build a date
`YEAR / MONTH / DAY`	Parts of a date
`WEEKDAY(d, 2)`	1 = Monday … 7 = Sunday
`EOMONTH(d, n)`	Month end n months away
`EDATE(d, n)`	Same day n months away
`NETWORKDAYS / WORKDAY`	Business days

⚠ COMMON MISTAKES

- Dates stored as text (left-aligned); date math fails. Use DATEVALUE or Text to Columns
- Joining a date with `&` and getting 46085; wrap it in `TEXT(date, "m/d/yyyy")`
- Using TEXT for numbers you still need to calculate with; it returns text

✓ TIPS & BEST PRACTICES

- Use TEXTBEFORE / TEXTAFTER instead of LEFT + FIND chains
- Keep a Holidays table and pass it to NETWORKDAYS / WORKDAY
- Test regex patterns on a few cells with REGEXTEST before replacing

QUICK SUMMARY

- `TEXTSPLIT / TEXTBEFORE / TEXTAFTER` — split
- `TEXTJOIN` — join with delimiter
- `REGEXTEST / EXTRACT / REPLACE` — patterns
- `EOMONTH / NETWORKDAYS` — date math

LET, LAMBDA & GROUPBY

FORMULAS THAT READ LIKE CODE

LET names intermediate results inside a formula, so a long calculation is readable and each part is computed once. LAMBDA turns a formula into your own function; save it in **Name Manager** and call it like a built-in. GROUPBY and PIVOTBY build summary tables with one formula, a live alternative to a PivotTable.

NOTES

- `LET(name1, value1, name2, value2, …, result)`
- Named LAMBDA: Formulas › Name Manager › New › Refers to: `=LAMBDA(x, …)`
- `BYROW(range, LAMBDA(r, …))` applies a calculation to each row
- GROUPBY / PIVOTBY / TRIMRANGE need a current Microsoft 365 build
- `A:.A` (trim-ref) means column A trimmed to used cells, like `TRIMRANGE(A:A)`

LET, LAMBDA & SUMMARIES

```
=LET(total, SUM(Sales[Amount]),
     west,  SUMIFS(Sales[Amount], Sales[Region], "West"),
     west / total)                                  # West share

# Name Manager → Name: TAXED   Refers to:
=LAMBDA(amount, rate, ROUND(amount*(1+rate), 2))
=TAXED(B2, 0.0825)                                  # → 108.25 for 100

=BYROW(B2:E20, LAMBDA(r, MAX(r)-MIN(r)))          # range per row
=MAP(A2:A20, LAMBDA(x, PROPER(TRIM(x))))          # clean each cell
=REDUCE(0, C2:C20, LAMBDA(acc, x, acc + (x>100)))  # count > 100

=GROUPBY(Sales[Region], Sales[Amount], SUM)       # totals by region
=PIVOTBY(Sales[Region], Sales[Year], Sales[Amount], SUM)  # cross-tab
=SUM(C2:.C10000)                                  # trims empty rows
```

CODE EXPLAINED

- → Formulas can span lines: `⌥↩` / `Alt+Enter` in the formula bar; LET reads best that way
- → A LAMBDA typed in a cell with arguments after it runs once: `=LAMBDA(x, x*2)(5)` → 10
- → GROUPBY takes the function by name (`SUM` , `AVERAGE` , `COUNTA`) or a LAMBDA
- → The dot in `C2:.C10000` trims trailing blank rows so whole-column formulas stay fast

OTHER SUITES

- ⇄ **Google Workspace** — LET, LAMBDA, MAP, BYROW, REDUCE, Named functions; QUERY instead of GROUPBY
- ⇄ **Apple iWork** — recent Numbers has LET, LAMBDA and SCAN; no GROUPBY / PIVOTBY (use a pivot table)

PLATFORM DIFFERENCES

- ▣ **Windows / Mac / Web** — LET and LAMBDA in M365 and 2024; GROUPBY / PIVOTBY / TRIMRANGE M365 only

LAMBDA HELPERS

FUNCTION	DOES
`MAP(arr, λ)`	Same-size result, item by item
`BYROW / BYCOL(arr, λ)`	One result per row / column
`REDUCE(init, arr, λ)`	Accumulate to one value
`SCAN(init, arr, λ)`	Running totals (spills)
`MAKEARRAY(r, c, λ)`	Build a grid
`ISOMITTED(arg)`	Optional LAMBDA arguments

GROUPBY ARGUMENTS

ARG	MEANING
`row_fields`	Group by these columns
`values`	Columns to aggregate
`function`	SUM, AVERAGE, COUNTA, PERCENTOF…
`field_headers`	3 = show headers
`total_depth`	0 none, 1 grand total, 2 subtotals
`sort_order`	-2 = by 2nd column descending

⚠ COMMON MISTAKES

- Naming a LET variable like a cell (`A1` , `R2C2`); use words like `rate`
- Defining LAMBDAs in one workbook and expecting them elsewhere; copy a sheet that uses them
- Sharing GROUPBY files with Office 2021 users, who see #NAME?

✓ TIPS & BEST PRACTICES

- Refactor long formulas with LET first; it's easier to debug one named step at a time
- Add comments to named LAMBDAs in Name Manager so others know the arguments
- Use GROUPBY for summaries that must update without clicking Refresh

QUICK SUMMARY

- `LET` — name steps inside a formula
- `LAMBDA` — custom functions
- `BYROW / MAP / REDUCE` — loop over ranges
- `GROUPBY / PIVOTBY` — formula PivotTables

PivotTables & Charts

Insert PivotTable · Rows / Columns / Values / Filters · group dates · Show Values As · calculated fields · PivotCharts · sparklines

SUMMARIES WITHOUT FORMULAS

A **PivotTable** summarises thousands of rows by dragging fields into **Rows**, **Columns**, **Values** and **Filters**. It groups dates by month or quarter, shows values as percentages, and drills into detail with a double-click. **PivotCharts** chart the result; **Recommended Charts** suggests a chart for any range; **sparklines** are tiny charts inside cells.

NOTES

- Source should be a table (sheet 16) so Refresh picks up new rows
- Refresh: `Alt+F5`, Refresh All: `Ctrl+Alt+F5` (Mac: PivotTable Analyze / Data › Refresh All)
- Right-click a value › **Show Values As** › % of Grand Total, Running Total, Rank…
- Double-click any value to see the rows behind it on a new sheet
- Insert chart from selection: `⌥F1` / `Alt+F1` embedded, `F11` new chart sheet

AUTOMATE IT

- Data › Queries & Connections › Properties › **Refresh data when opening the file**
- One-line VBA event: `Private Sub Workbook_Open(): ThisWorkbook.RefreshAll: End Sub`

PIVOT & CHART KEYS

```
Insert › PivotTable › From Table/Range     # Alt N V T (Win)
Drag fields:  Rows = Region   Columns = Year   Values = Sum of Amount

Analyze › Refresh  / Alt+F5        # refresh this PivotTable
Data › Refresh All / Ctrl+Alt+F5   # refresh all
⌥F1                / Alt+F1        # chart from selection (same sheet)
F11                / F11           # chart on its own sheet

=GETPIVOTDATA("Amount", $A$3, "Region", "West")  # read a pivot cell
Right-click date › Group › Months + Years        # date grouping
```

CODE EXPLAINED

- → Clicking a pivot cell while writing a formula creates GETPIVOTDATA; turn it off in PivotTable › Options
- → Date grouping usually happens automatically; regroup via right-click › Group
- → Calculated fields (PivotTable Analyze › Fields, Items & Sets) work on sums, not row by row
- → PivotCharts follow the pivot's filters and slicers

OTHER SUITES

- ⇄ **Google Workspace** — Insert › Pivot table, suggested pivots, Chart editor
- ⇄ **Apple iWork** — Numbers pivot tables (Organize › Create Pivot Table) and charts

PLATFORM DIFFERENCES

- ▣ **Windows** — Data Model, Power Pivot, calculated items
- ▣ **Mac** — PivotTables and slicers; no Power Pivot
- ▣ **Web** — create / edit PivotTables, fewer options

PIVOT AREAS

AREA	PUT HERE
Rows	Categories down the side
Columns	Categories across the top (few values)
Values	Numbers to sum / count / average
Filters	Report-level filter
Slicer / Timeline	Click-to-filter buttons / date range

PICK A CHART

SHOW	CHART
Trend over time	Line
Compare categories	Clustered bar / column
Part of a whole (≤5)	Stacked bar (or pie)
Two measures	Combo with secondary axis
Distribution	Histogram / box & whisker
Trend in a cell	Sparkline (Insert › Sparklines)

⚠ COMMON MISTAKES

- Building pivots on a fixed range, then adding rows that never appear; use a table
- Forgetting to refresh; pivots don't update when source data changes
- Count instead of Sum because the column has text or blanks; fix the data and pick Sum

✓ TIPS & BEST PRACTICES

- Use Tabular layout + Repeat All Item Labels for pivots you'll copy elsewhere
- Add a Timeline slicer for date filtering
- Turn off GETPIVOTDATA if you want plain cell references

QUICK SUMMARY

- `Rows / Columns / Values / Filters` — pivot areas
- `Alt+F5` — refresh
- `Show Values As` — % and running totals
- `Alt+F1` / `F11` — quick chart

Sheet 23

Conditional Formatting & Data Tools

rule types · formula-based rules · Remove Duplicates · Text to Columns · sort & filter ⌘⇧F / Ctrl+Shift+L · Goal Seek · number formats

HIGHLIGHT PROBLEMS, FIX THEM IN BULK

Conditional formatting colours cells by rule: above average, duplicates, data bars, or any formula. **Data tools** fix messy data in bulk: **Remove Duplicates**, **Text to Columns**, sort and filter. **Goal Seek** works backwards from a target ("what price gives $10,000 profit?"). **Custom number formats** change how values look without changing them.

NOTES

- Formula rules are written for the **top-left cell** of the selection with relative refs
- Highlight a whole row: select A2:F100 › rule `=$E2="Overdue"`
- Manage Rules shows rule order; tick Stop If True to prevent overlaps
- Filter: `⌘⇧F` (Mac) / `Ctrl+Shift+L`; open a column's filter with `⌥↓` / `Alt+↓`
- Goal Seek: Data › What-If Analysis › Goal Seek

AUTOMATE IT

- Office Script (sheet 35) that trims, dedupes and formats each new export the same way
- Keep rules on a table so they extend to new rows

RULES & FORMATS

```
# Conditional formatting formulas (Home › Conditional Formatting › New Rule › Use a formula)
=$E2="Overdue"                      # whole row red when E says Overdue
=AND($D2<TODAY(), $F2<>"Done")      # past due and not done
=COUNTIF($A$2:$A$500, $A2)>1        # duplicates in column A
=MOD(ROW(), 2)=0                    # banded rows
=$C2>AVERAGE($C$2:$C$500)           # above average

# Custom number formats (⌘1 / Ctrl+1 › Number › Custom)
#,##0;[Red]-#,##0;"—"              # positive; negative red; zero as dash
0.0,,"M"                            # 2500000 → 2.5M
"Qty: "0                            # 5 → Qty: 5
[<1000]0;[<1000000]0.0,"K";0.0,,"M"  # auto K / M
```

CODE EXPLAINED

- → `$E2` locks the column but not the row, so each row checks its own E cell
- → Number format sections: `positive;negative;zero;text`
- → Each comma after the digits divides by 1,000 in the display only
- → `⌘1` / `Ctrl+1` opens Format Cells from anywhere

OTHER SUITES

- ⇄ **Google Workspace** — Format › Conditional formatting › Custom formula; Data cleanup tools
- ⇄ **Apple iWork** — Numbers Conditional Highlighting (no formula rules); custom formats in the Format sidebar

PLATFORM DIFFERENCES

- ▣ **Windows** — every rule type, Data Tables, Solver
- ▣ **Mac** — same rules; Solver add-in available
- ▣ **Web** — formula rules, Remove Duplicates; no Goal Seek

DATA TOOLS

TOOL	USE
Remove Duplicates	Delete repeat rows (by chosen columns)
Text to Columns	Split by delimiter / fixed width; fix text dates
Sort (custom)	Several levels, custom lists
Advanced Filter	Copy unique records elsewhere
Goal Seek	One input, one target
Data Table	Grid of results for two inputs
Solver	Optimise with constraints (add-in)

RULE TYPES

RULE	EXAMPLE
Highlight Cells	> 100, text contains, dates
Top / Bottom	Top 10, above average
Data Bars	In-cell bar length
Color Scales	Heat map
Icon Sets	Arrows / traffic lights
Formula	Anything TRUE / FALSE

⚠ COMMON MISTAKES

- Writing a row rule as `=E2="Overdue"` without `$`, so every column checks a different cell
- Running Remove Duplicates on a copy you didn't keep; it deletes rows permanently
- Changing number formats and thinking the value changed; rounding is display only

✓ TIPS & BEST PRACTICES

- Select the whole data area before creating a rule, then write it for the first row
- Copy the data to a new sheet before big cleanups
- Use `Ctrl+Shift+L` twice to clear all filters at once

QUICK SUMMARY

- `=$E2="x"` — row-based rule
- `⌘⇧F` / `Ctrl+Shift+L` — filter on / off
- `pos;neg;zero;text` — number format sections
- `Goal Seek` — work back from a target

Power Query

RECORD YOUR CLEANUP ONCE

Power Query (Data › Get Data) imports from files, folders, databases and the web, then records every cleanup step you click: remove columns, split, change types, unpivot, merge. Next month, drop in the new file and click **Refresh All**: every step replays. It replaces copy-paste-and-fix routines with a repeatable recipe.

NOTES

- Each click adds an **Applied Step**; select a step to see the data at that point
- **From Folder** combines every file in a folder with the same layout
- **Merge** = join (like XLOOKUP); **Append** = stack tables
- **Unpivot Columns** turns month columns into Month + Value rows
- Close & Load To › **Only Create Connection** for staging queries

AUTOMATE IT

- Power Query replaces manual monthly imports: point it at a folder, Refresh All each month
- Refresh on open (Query Properties) so reports are always current

TYPICAL STEPS (M CODE)

```
let
    Source   = Excel.Workbook(File.Contents("C:\Data\sales.xlsx")),
    Sheet    = Source{[Item="Sales",Kind="Sheet"]}[Data],
    Headers  = Table.PromoteHeaders(Sheet),
    Types    = Table.TransformColumnTypes(Headers,
                 {{"Date", type date}, {"Amount", type number}}),
    NoBlanks = Table.SelectRows(Types, each [Amount] <> null),
    Unpivot  = Table.UnpivotOtherColumns(NoBlanks, {"Region"}, "Month", "Value"),
    Trimmed  = Table.TransformColumns(Unpivot, {{"Region", Text.Trim}})
in
    Trimmed
```

CODE EXPLAINED

- → You rarely type M; every ribbon click writes a step like these
- → Each step references the previous one by name, so steps can be renamed and reordered
- → `each [Amount] <> null` is a row filter; `[Column]` refers to the current row
- → Change the source path in one place (Source step or a parameter) for a new file

OTHER SUITES

- ⇄ **Google Workspace** — no Power Query; use IMPORTRANGE, QUERY, Apps Script or Connected Sheets
- ⇄ **Apple iWork** — no equivalent in Numbers; prepare data elsewhere or script it

PLATFORM DIFFERENCES

- ▣ **Windows** — full editor, every connector
- ▣ **Mac** — Power Query editor available in M365, fewer connectors
- ▣ **Web** — refresh and some editing; limited sources

COMMON TRANSFORMATIONS

BUTTON	DOES
`Remove Columns / Choose Columns`	Drop what you don't need
`Split Column`	By delimiter or position
`Change Type`	Text, number, date (do it early)
`Fill Down`	Fill blanks from above
`Group By`	Summaries
`Unpivot / Pivot`	Columns ↔ rows
`Replace Values / Trim / Clean`	Fix text

MERGE JOIN KINDS

KIND	KEEPS
`Left outer`	All left rows + matches (like XLOOKUP)
`Inner`	Only rows in both
`Full outer`	Everything
`Left anti`	Left rows with no match
`Right outer / anti`	Mirror of the above

⚠ COMMON MISTAKES

- Changing types too late, after steps that already depended on text values
- Hard-coded file paths that break on a coworker's PC; use a parameter or SharePoint path
- Loading every staging query to a sheet; use Connection Only

✓ TIPS & BEST PRACTICES

- Name steps (right-click › Rename) so the recipe reads like a checklist
- Use From Folder for monthly files; new files are picked up on refresh
- Check the column quality bar (View › Column quality) for errors and empties

QUICK SUMMARY

- `Get Data` — import and record steps
- `Merge / Append` — join / stack
- `Unpivot` — columns to rows
- `Refresh All` — replay everything

Slide Master & Layouts

DESIGN ONCE, EVERY SLIDE FOLLOWS

The **Slide Master** is the top slide that sets fonts, colours, background and logo for the whole deck. Under it sit **layouts** (Title, Title and Content, Two Content…), each with **placeholders** for text and pictures. Fix the design in the master and every slide updates; fix it on individual slides and you're fixing it forever.

NOTES

- Open: View › Slide Master; close with **Close Master View**
- Edit the top (master) slide for deck-wide changes; edit a layout for that layout only
- Insert Placeholder adds picture, chart or text boxes to a custom layout
- Footers and slide numbers: Insert › Header & Footer › tick Slide number
- **Reset** (Home tab) snaps a slide back to its layout after manual edits

AUTOMATE IT

- Save the master as a `.potx` template (sheet 04) so new decks start on-brand
- Use layouts with picture placeholders so pasted images crop to the right size

BUILD A CLEAN MASTER

1. View › **Slide Master**; select the top master slide
2. Set theme **Colors** and **Fonts** (Slide Master tab) instead of formatting text boxes
3. Add the logo and background to the master; they appear on every layout
4. Delete layouts you'll never use; rename the rest (right-click › Rename Layout)
5. Create custom layouts with Insert Placeholder for repeated slide types
6. Close Master View, then apply layouts with Home › **Layout**
7. File › Save As › **PowerPoint Template (.potx)**

WHERE TO FIND IT

TASK	WHERE
`Slide Master`	View › Slide Master
`Change a slide's layout`	Home › Layout
`Reset slide`	Home › Reset
`Slide numbers / footer`	Insert › Header & Footer
`Slide size 16:9 / 4:3`	Design › Slide Size
`Apply another deck's theme`	Design › Themes › Browse

OTHER SUITES

- ⇄ **Google Workspace** — Slide › Edit theme (master + layouts); import theme from another deck
- ⇄ **Apple iWork** — Keynote: Edit Master Slides; Save Theme to reuse

PLATFORM DIFFERENCES

- **Windows** — full master editing
- **Mac** — same features
- **Web** — apply layouts and themes; Slide Master editing limited

LAYOUTS WORTH KEEPING

LAYOUT	USE FOR
`Title Slide`	Opening slide
`Section Header`	Divides parts of the talk
`Title and Content`	Most slides
`Two Content / Comparison`	Side by side
`Title Only`	Charts and diagrams
`Blank`	Full-bleed images

MASTER VS LAYOUT VS SLIDE

EDIT ON	CHANGES
`Master (top)`	Every layout and slide
`Layout`	Slides using that layout
`Slide`	That slide only (avoid)
`Theme`	Colors + fonts everywhere

⚠ COMMON MISTAKES

- Pasting the logo on every slide; it belongs on the master
- Typing in free text boxes instead of placeholders, so text isn't in the outline or accessibility order
- Changing fonts slide by slide; change theme fonts once

✓ TIPS & BEST PRACTICES

- Hold `⇧` while clicking Normal view (Mac) / the Normal button (Win) to jump into the master
- Use Reset after pasting slides from another deck to adopt your layouts
- Keep a hidden "style guide" slide in templates showing approved colours

QUICK SUMMARY

- `View › Slide Master` — deck-wide design
- `Layouts` — reusable slide shapes
- `Reset` — back to the layout
- `.potx` — save as template

Building Slides Fast

FROM OUTLINE TO FINISHED DECK

Write the talk in **Outline view** (every line is a slide title or bullet), then let **Designer** suggest layouts for each slide. **Duplicate** a finished slide instead of starting from blank. **Align & Distribute**, **Smart Guides** and the **Selection Pane** make objects line up and keep busy slides manageable.

NOTES

- Outline view: View › Outline; `Tab` demotes a title to a bullet, `⇧Tab` promotes
- Designer (Home › Designer) suggests layouts when you add pictures or lists
- Reuse Slides (Home › New Slide › Reuse Slides) pulls slides from another deck
- Selection Pane: rename, hide and reorder overlapping objects
- Nudge with arrow keys; hold `⌥` / `Ctrl` for fine nudges

AUTOMATE IT

- Reuse Slides from a library deck instead of rebuilding standard slides
- Designer applies a consistent look to pictures and lists in one click

BUILDING SHORTCUTS

```
⌘D              / Ctrl+D          # duplicate slide or object
⇧⌘N / ^M        / Ctrl+M          # new slide
⌥⌘G             / Ctrl+G          # group objects
⇧⌥⌘G            / Ctrl+Shift+G    # ungroup
⇧ + drag                          # constrain move / keep proportions
⌥ + drag        / Ctrl + drag     # drag a copy
⌘] ⌘[           / Ctrl+] Ctrl+[   # bigger / smaller text
Arrange menu    / Arrange menu    # bring to front / send to back
Arrange › Sel.  / Alt+F10         # Selection Pane
⌘1              / Ctrl+1 (Format) # format pane
```

CODE EXPLAINED

- → `⌘D` / `Ctrl+D` on a selected object makes a copy offset by the last move, so repeating it spaces evenly
- → Arrange › Align works on the selection; tick **Align Selected Objects** vs **Align to Slide**
- → Distribute Horizontally needs three or more objects
- → Ordering shortcuts differ by version; Arrange menu always works

OTHER SUITES

- ⇄ **Google Workspace** — `⌘D` duplicate, Arrange › Align & distribute, `⌘⌥G` group
- ⇄ **Apple iWork** — Keynote: alignment guides, Arrange › Align / Distribute, ⌥-drag to copy

PLATFORM DIFFERENCES

- ▣ **Windows** — Designer, Selection Pane `Alt+F10`, Reuse Slides
- ▣ **Mac** — same; Selection Pane under Arrange
- ▣ **Web** — Designer, align tools; Reuse Slides limited

ARRANGE TOOLS

TOOL	USE
`Align Left / Center / Right`	Line up edges
`Align Top / Middle / Bottom`	Line up vertically
`Distribute H / V`	Equal gaps (3+ objects)
`Group`	Move as one
`Selection Pane`	Pick hidden / stacked objects
`Smart Guides`	Red lines while dragging

OUTLINE VIEW KEYS

KEY	DOES
`Enter`	New item at same level
`Tab`	Demote (slide → bullet)
`⇧Tab`	Promote (bullet → new slide)
`⌥⇧↑ ↓ / Alt+Shift+↑ ↓`	Move item / slide
`^⇧1 / Alt+Shift+1`	Show titles only

⚠ COMMON MISTAKES

- Eyeballing alignment; a few pixels off is visible when projected
- Building slides from Blank layout with text boxes, losing outline and accessibility order
- Piling objects on top of each other with no names; use the Selection Pane

✓ TIPS & BEST PRACTICES

- Draft the whole talk in Outline view first, then design
- Turn on View › Guides for consistent margins
- Copy formatting with Format Painter or `⇧⌘C` / `Ctrl+Shift+C`

QUICK SUMMARY

- `⌘D` / `Ctrl+D` — duplicate
- `Outline view` — write first, design later
- `Align & Distribute` — tidy objects
- `Selection Pane` — manage layers

Animations, Morph & Zoom

MOTION THAT EXPLAINS

Morph is the most useful transition: duplicate a slide, move or resize things on the copy, and Morph animates the change smoothly. **Animations** bring objects in, out or along a path; the **Animation Pane** controls order and timing. **Zoom** (Insert › Zoom) makes a clickable overview slide that jumps to sections and back.

NOTES

- Morph matches objects by type and content; name them `!!logo` in the Selection Pane to force a match
- Animation types: **Entrance**, **Emphasis**, **Exit**, **Motion Path**
- Start: On Click, With Previous, After Previous
- **Animation Painter** copies an animation to other objects
- **Triggers** start an animation when a specific object is clicked

USE MORPH

1. Finish slide A with the objects in their start positions
2. Duplicate it (`⌘D` / `Ctrl+D`) to make slide B
3. On slide B, move, resize, recolour or rotate the objects
4. Select slide B › Transitions › **Morph**
5. Effect Options › Objects, Words or Characters
6. Play from slide A (`⇧⌘↩` / `Shift+F5`) to preview

WHERE TO FIND IT

TASK	WHERE
`Morph`	Transitions › Morph
`Animation Pane`	Animations › Animation Pane
`Copy animation`	Animations › Animation Painter
`Trigger`	Animations › Trigger › On Click of
`Summary Zoom`	Insert › Zoom › Summary Zoom
`Apply to all slides`	Transitions › Apply To All

OTHER SUITES

⇄ **Google Workspace** — Insert › Animation; transitions; no Morph or Zoom

⇄ **Apple iWork** — Keynote **Magic Move** (same idea as Morph), builds, Action builds

PLATFORM DIFFERENCES

- **Windows** — Morph, Zoom, triggers, motion paths
- **Mac** — Morph and Zoom supported
- **Web** — plays Morph and Zoom; editing animations is limited

START OPTIONS

OPTION	PLAYS
`On Click`	Next click or →
`With Previous`	Same time as the one before
`After Previous`	When the one before ends
`Delay`	Wait before starting
`Duration`	How long the effect runs

ZOOM TYPES

ZOOM	CREATES
`Summary Zoom`	Overview slide of chosen sections
`Section Zoom`	Link to a section, returns after
`Slide Zoom`	Link to one slide inside another
`Return to Zoom`	Option that brings you back

⚠ COMMON MISTAKES

- Animating every bullet on every slide; it slows the talk
- Morph between slides with differently named objects, so things fade instead of move
- Long durations; most effects read best at 0.3–0.5 s

✓ TIPS & BEST PRACTICES

- Use Morph for "zoom in on part of a chart" by scaling the chart up on slide B
- Prefix shared object names with `!!` to make Morph pair them
- Use Summary Zoom as a non-linear menu slide for Q&A

QUICK SUMMARY

- `Morph` — duplicate, change, transition
- `Animation Pane` — order and timing
- `Triggers` — click an object to animate
- `Zoom` — jump to sections

Sheet 28

Presenting & Exporting

start ⇧⌘↩ / F5 · from current slide · Presenter View · B / W blank · number + Enter · Rehearse Timings · Speaker Coach · subtitles · export video

RUN THE ROOM FROM THE KEYBOARD

Presenter View shows your notes, the next slide and a timer on your screen while the audience sees only the slide. During a show, the keyboard does everything: jump to any slide by number, blank the screen to pull attention back, draw on a slide. **Rehearse Timings** and **Speaker Coach** help you practise; **Export** turns a deck into PDF or video.

NOTES

- Slide Show › **Use Presenter View**; swap screens with Display Settings › Swap
- Type a slide number and press Enter to jump straight to it
- `B` / `W` blanks to black / white; press again to return
- Live subtitles: Slide Show › Subtitle Settings (translate while you talk)
- Speaker Coach (Slide Show › Rehearse with Coach) flags pace, fillers and reading slides

SLIDE SHOW KEYS

```
⇧⌘↩          / F5              # start from the beginning
⌘↩           / Shift+F5        # start from current slide
→ ↓ Space N  /                 # next
← ↑ P        /                 # previous
12 ↩         / 12 Enter        # go to slide 12
B   W        /                 # black / white screen
⌘P  ⌘E       / Ctrl+P  Ctrl+E  # pen / eraser (Win: laser Ctrl+L)
E            /                 # erase all ink
—            / Ctrl+S          # all slides list (Windows)
Esc          / Esc             # end show
?  or  F1                      # show the shortcut list during a show
```

CODE EXPLAINED

- → Keys are the same during any show; `F1` (Win) or `?` lists them on screen
- → Ink you draw can be kept or discarded when the show ends
- → Slides marked **Hide Slide** are skipped unless you jump to them by number
- → Rehearse Timings records time per slide for kiosk or auto-advance shows

OTHER SUITES

- ⇄ **Google Workspace** — Present `⌘↩` / `Ctrl+Enter`, Presenter view, captions `⌘⇧C`, laser `L`
- ⇄ **Apple iWork** — Keynote Play `⌥⌘P`, presenter display, `X` swap displays, `B` / `W`

PLATFORM DIFFERENCES

- ▣ **Windows** — Presenter View, Speaker Coach, Cameo, record to video
- ▣ **Mac** — Presenter View, recording; some coach features later
- ▣ **Web** — present, Speaker Coach, subtitles; no recording

EXPORT OPTIONS

EXPORT	WHERE
PDF / handouts	File › Export › PDF
Video (MP4)	File › Export › Create a Video
Images per slide	Save As › PNG / JPEG
Record with narration	Record tab › Record
Package for CD / folder	Export › Package Presentation (Win)

BEFORE YOU PRESENT

CHECK	WHY
Embed fonts	Options › Save › Embed fonts
Compress media	File › Info › Compress Media
Presenter View on	Notes on your screen only
Turn off notifications	Focus / Do Not Disturb
Run from local copy	No Wi-Fi surprises

⚠ COMMON MISTAKES

- Clicking back through 20 slides to answer a question; type the number and press Enter
- Presenting from a cloud file over weak Wi-Fi; download it first
- Mirroring displays, so the audience sees your notes

✓ TIPS & BEST PRACTICES

- Print or memorise slide numbers of your backup slides for Q&A
- Press `B` to black out the screen during discussion
- Export a PDF backup in case the venue PC lacks your fonts

QUICK SUMMARY

- `F5` / `⇧⌘↩` — start show
- `12 Enter` — jump to slide 12
- `B / W` — blank screen
- `Presenter View` — notes on your screen

Outlook Mail Efficiency

new Outlook vs classic · Focused Inbox · search operators · flags & categories · Sweep · Undo Send · Schedule Send · reading shortcuts

PROCESS MAIL, DON'T JUST READ IT

Outlook has three faces: **classic Outlook** for Windows, the **new Outlook** (same app as Outlook on the web), and **Outlook for Mac**. All of them support **Focused Inbox**, **search operators**, **flags**, **categories** and **Sweep** (bulk-clean a sender). The goal is to touch each message once: reply, file, flag or delete.

NOTES

- Search operators: `from:` `to:` `subject:` `hasattachments:yes` `received:last week`
- **Sweep** (web / new Outlook): delete or move all mail from a sender, now and in future
- **Undo Send**: Settings › Mail › Compose and reply › delay up to 10 seconds
- Schedule Send: Send dropdown › Schedule send
- Flags create tasks in To Do; categories colour-code across mail and calendar

MAIL SHORTCUTS

```
⌘N            / Ctrl+N           # new message
⌘R  ⇧⌘R       / Ctrl+R  Ctrl+Shift+R   # reply / reply all
⌘J            / Ctrl+F           # forward
⌘↩            / Ctrl+Enter       # send
⌘⌥F / ⌥⌘F     / Ctrl+E           # search
⇧⌘M           / Ctrl+Shift+V     # move to folder
⌫             / Delete           # delete;  ^E / Backspace = archive (new Outlook)
⌘T            / Ctrl+Q           # mark read  (Ctrl+U unread)
⌘'            / Insert           # flag (classic Win)
⌘1  ⌘2        / Ctrl+1  Ctrl+2   # mail / calendar

from:"Dana Lee" hasattachments:yes received:>=1/1/2026
```

CODE EXPLAINED

- → Shortcuts differ between classic Windows and new Outlook / web; Settings › General › Keyboard shortcuts can switch the web to Gmail-style keys
- → Search operators can be combined; use quotes for names with spaces
- → `Ctrl+F` forwards in Outlook (not Find); search is `Ctrl+E`
- → Archive keeps the inbox empty without deleting

OTHER SUITES

- ⇄ **Google Workspace** — Gmail search operators, labels, snooze, schedule send, undo send
- ⇄ **Apple iWork** — Apple Mail Smart Mailboxes, VIPs, Send Later, Remind Me

PLATFORM DIFFERENCES

- ▣ **Windows (classic)** — Quick Steps, rules, VBA, offline PST files
- ▣ **New Outlook / Web** — Sweep, pinning, Gmail-style keys option
- ▣ **Mac** — same core; some shortcuts are Mac-specific

SEARCH OPERATORS

OPERATOR	FINDS
`from: / to: / cc:`	People
`subject:budget`	Word in subject
`hasattachments:yes`	With attachments
`attachments:report.xlsx`	Attachment name
`received:yesterday`	Date (also `last week`, `>=1/1/2026`)
`is:unread / is:flagged`	State (new Outlook / web)
`category:Red`	Category

TOUCH IT ONCE

MESSAGE	ACTION
`< 2 minutes`	Reply now, archive
`Needs work later`	Flag (due date) or move to To Do
`Reference`	Archive (search finds it)
`Waiting on others`	Category "Waiting"
`Newsletters`	Sweep or unsubscribe

⚠ COMMON MISTAKES

- Hitting `Ctrl+F` to search and forwarding the email instead
- Building deep folder trees; search plus a few folders is faster
- Leaving read mail in the inbox as a to-do list; flag it so it shows in To Do

✓ TIPS & BEST PRACTICES

- Turn on Focused Inbox and train it with "Move to Other"
- Use categories that match projects, so mail and meetings share the colour
- Set Undo Send to 10 seconds

QUICK SUMMARY

- `from: subject: hasattachments:` — search
- `Ctrl+R / Ctrl+Shift+R` — reply / all
- `Sweep` — bulk-clean a sender
- `Flag` — becomes a task

Sheet 30

Rules, Quick Steps & Templates

Rules · Quick Steps · My Templates · signatures · Quick Parts in mail · automatic replies · conditional formatting

LET OUTLOOK DO THE FILING

Rules act on mail automatically as it arrives (move newsletters, flag mail from your boss). **Quick Steps** bundle several actions into one click or shortcut (forward to team, categorise and archive). **Templates** and **Quick Parts** store messages you send repeatedly. **Automatic replies** handle out-of-office.

NOTES

- Create a rule from a message: right-click › Rules › Create Rule
- Server-side rules run even when Outlook is closed; client-only rules don't
- Quick Steps (classic Win / new Outlook): Home › Quick Steps; assign `Ctrl+Shift+1…9`
- My Templates (web / new): compose › ··· › My Templates; classic: Save as `.oft`
- Automatic replies: File › Automatic Replies (classic) or Settings › Mail › Automatic replies

AUTOMATE IT

- Rules for anything you file the same way more than twice
- Quick Steps for multi-step actions like "reply with template, categorise, archive"

SET UP A RULE AND A QUICK STEP

1. Right-click a newsletter › **Rules › Create Rule**
2. Condition: From sender; Action: Move to folder "Newsletters"
3. Tick **Run this rule now on messages already in the current folder**
4. Home › Quick Steps › **Create New**; name it "To Team"
5. Add actions: Forward to team@..., Categorise "Shared", Move to Archive
6. Pick a shortcut key (`Ctrl+Shift+1`) and Finish

WHERE TO FIND IT

TASK	WHERE
Rules	Settings › Mail › Rules / File › Manage Rules
Quick Steps	Home › Quick Steps
Templates	Compose › ··· › My Templates
Signatures	Settings › Mail › Compose › Signatures
Out of office	Settings › Automatic replies
Highlight mail	View settings › Conditional Formatting

OTHER SUITES

- ⇄ **Google Workspace** — Gmail filters (from search), templates, vacation responder
- ⇄ **Apple iWork** — Apple Mail rules (Settings › Rules), signatures per account

PLATFORM DIFFERENCES

- **Windows (classic)** — every rule option, Quick Steps, `.oft` templates, VBA rules
- **New Outlook / Web** — server rules, My Templates, Quick Steps
- **Mac** — rules, signatures; no Quick Steps

USEFUL RULES

WHEN	DO
From newsletter senders	Move to Newsletters
Sent only to me + from manager	Flag / category Important
Subject has [JIRA]	Move to Tickets, mark read
I'm in CC only	Move to CC folder
Has attachment from client	Category Client

REUSABLE TEXT

TOOL	BEST FOR
Signatures	Name / title block, one per account
My Templates	Whole messages you resend
Quick Parts (classic)	Paragraphs inserted mid-message
Suggested replies	Short AI replies (new Outlook)

⚠ COMMON MISTAKES

- Rules that run "on this computer only", so mail isn't sorted on your phone
- Overlapping rules without **Stop processing more rules**, moving mail twice
- Out-of-office without an end date, still replying a month later

✓ TIPS & BEST PRACTICES

- Order rules most specific first
- Export rules (classic: Manage Rules › Options › Export) as a backup
- Set separate internal and external automatic replies

QUICK SUMMARY

- `Rules` — automatic filing
- `Quick Steps` — multi-action buttons
- `My Templates` — saved messages
- `Automatic replies` — out of office

Calendar & Scheduling

Scheduling Assistant · room finder · shared calendars · time zones · Bookings · view shortcuts · drag mail to calendar

FIND TIME WITHOUT EMAIL PING-PONG

The **Scheduling Assistant** shows attendees' free / busy side by side so you pick a slot that works. **Shared calendars** let you see a team's availability. **Bookings** gives people a page to book time with you. Dragging an email onto the calendar turns it into an appointment with the message attached.

NOTES

- New event: `⌘N` in Calendar / `Ctrl+Shift+Q` (meeting request) on Windows
- Add a second time zone: Settings › Calendar › Time zones
- Open a coworker's calendar: Add calendar › From directory
- Teams meeting toggle adds a join link to any event
- **Bookings** (business plans) or a personal booking page for 1:1s

AUTOMATE IT

- A Bookings page instead of trading three emails per meeting
- Scheduling Assistant suggestions (Suggested times) instead of manual comparison

CALENDAR SHORTCUTS

```
⌘2                / Ctrl+2           # go to calendar
⌘N                / Ctrl+Shift+Q     # new meeting / event
^⌘1 … ^⌘4         / Ctrl+Alt+1 … 4   # day / work week / week / month
⌘T                / Alt+Home or T    # today
→ ←               / Alt+↓ ↑          # next / previous period (varies)
⌘O                / Ctrl+O           # open selected event
⌫                 / Ctrl+D           # delete event

# Natural entry in a new event's time box (web / new Outlook)
tomorrow 3pm     next fri 9am     in 2 weeks
```

CODE EXPLAINED

- → Classic Windows and new Outlook use slightly different view keys; hover buttons for hints
- → Ctrl+Shift+Q creates a meeting request with attendees; Ctrl+Shift+A an appointment
- → Natural-language dates work in the date fields of new Outlook and the web
- → Drag a message onto the Calendar icon to make an event from it

OTHER SUITES

- ⇄ **Google Workspace** — Google Calendar: find a time, appointment schedules, `c` new event
- ⇄ **Apple iWork** — Apple Calendar: shared iCloud calendars, natural-language quick event

PLATFORM DIFFERENCES

- ▣ **Windows (classic)** — Scheduling Assistant, room finder, VBA
- ▣ **New Outlook / Web** — Suggested times, Bookings integration
- ▣ **Mac** — Scheduling Assistant and room finder

SCHEDULING TOOLS

TOOL	USE
Scheduling Assistant	Free / busy grid
Room Finder	Free rooms by building
Suggested times	Auto-picked slots
Bookings / booking page	Others book you
FindTime / polls	Vote on options
Shared calendar	Team leave / on-call

CALENDAR HABITS

HABIT	SETTING
Shorter meetings	Settings › End appointments early
Focus time	Recurring blocks marked Busy
Working hours	Settings › Calendar › Work hours
Two time zones	Time zones › Add
Private event	Mark as Private

⚠ COMMON MISTAKES

- Booking over people's Tentative and Out of office blocks without checking the assistant
- Sharing your calendar with full details when free / busy would do
- Creating separate events for each time zone instead of using the time-zone picker

✓ TIPS & BEST PRACTICES

- Use "End appointments and meetings early" to get 5–10 minutes between meetings
- Colour events with the same categories as your email
- Block focus time as recurring Busy events so others see it

QUICK SUMMARY

- `Scheduling Assistant` — free / busy grid
- `Ctrl+Shift+Q` — new meeting
- `Bookings` — let people book you
- `Time zones` — show two side by side

Sheet 32

Teams, OneNote & Loop

Teams search ⌘E / Ctrl+E · mute ⌘⇧M / Ctrl+Shift+M · chat vs channels · meeting recap · OneNote tags · Loop components

THE REST OF THE SUITE

Teams is chat, channels and meetings; most time is saved with search, mute / camera shortcuts and meeting recaps. **OneNote** is the free-form notebook: tags turn notes into searchable to-dos. **Loop components** are live tables, lists and checklists you paste into a chat, email or doc; everyone edits the same copy.

NOTES

- Teams search box: `⌘E` / `Ctrl+E` ; `/` commands are mostly gone, use search
- Channels for topic history; chats for quick back-and-forth
- Meeting recap: recording, transcript, notes and (with Copilot) a summary
- OneNote tags: `⌘1` / `Ctrl+1` To Do, then search tags with Find Tags
- Loop components work in Teams chat, Outlook mail, Whiteboard and the Loop app

TEAMS & ONENOTE KEYS

```
# Teams
⌘E            / Ctrl+E            # search
⌘⇧M           / Ctrl+Shift+M      # mute / unmute
⌘⇧O           / Ctrl+Shift+O      # camera on / off
⌘⇧K           / Ctrl+Shift+K      # raise / lower hand
⌘⇧X           / Ctrl+Shift+X      # expand compose box
⌘.            / Ctrl+.            # show all shortcuts

# OneNote
⌘1            / Ctrl+1            # To Do tag
⌘2  ⌘3        / Ctrl+2  Ctrl+3    # Important / Question tags
⌘N            / Ctrl+N            # new page
⌘⌥F / ⌘F      / Ctrl+E            # search all notebooks
```

CODE EXPLAINED

→ Hold Space while muted in a Teams meeting to talk temporarily (if enabled)

→ `Ctrl+.` lists every Teams shortcut for your platform

→ OneNote tags are searchable across notebooks: Home › Find Tags (Windows)

→ A Loop table pasted into chat stays in sync wherever else it's pasted

OTHER SUITES

⇄ **Google Workspace** — Google Chat & Meet; Keep for notes; smart chips in Docs as live components

⇄ **Apple iWork** — FaceTime, Messages; Apple Notes with tags and checklists

PLATFORM DIFFERENCES

- **Windows** — Teams, OneNote desktop with every feature
- **Mac** — Teams and OneNote apps; same core features
- **Web** — Teams web, OneNote for the web, Loop app

WHICH APP?

NEED	USE
Quick question	Teams chat
Project discussion with history	Teams channel
Meeting notes	OneNote or Loop in the meeting
Shared checklist / table	Loop component
Personal research notebook	OneNote
Formal document	Word in SharePoint

MEETING HABITS

HABIT	HOW
Join muted	Pre-join screen toggles
Background blur	Effects › Blur
Live captions	··· › Language › Captions
Record + transcript	Record and transcribe
Agenda in invite	Loop agenda component

⚠ COMMON MISTAKES

- Long-running conversations in chats that should be channels; history gets lost
- Copying meeting notes into email; share the OneNote / Loop link
- Forgetting you're unmuted; learn `⌘⇧M` / `Ctrl+Shift+M`

✓ TIPS & BEST PRACTICES

- Pin your top channels and chats
- Use OneNote To Do tags + Find Tags for a weekly review
- Paste a Loop checklist in meeting chats for action items

QUICK SUMMARY

- `Ctrl+E` — Teams search
- `Ctrl+Shift+M` — mute
- `Ctrl+1` — OneNote To Do tag
- `Loop` — live shared components

Macros & VBA Basics

RECORD IT ONCE, REPLAY FOREVER

A **macro** is a recorded or written list of steps in **VBA** (Visual Basic for Applications). The **macro recorder** writes the code for you while you click; the **VBA editor** lets you clean it up. Macros live in macro-enabled files (`.xlsm`, `.docm`) or in the **Personal Macro Workbook**, which makes them available in every Excel file on your machine.

NOTES

- Turn on the Developer tab: Options › Customize Ribbon › tick Developer
- **Use Relative References** before recording if the macro should work from any cell
- Store in **Personal Macro Workbook** for macros you want everywhere (Excel)
- Run: Developer › Macros (`⌥F8` / `Alt+F8`), a shortcut key, a button or the QAT
- Macros from the internet are blocked by default; unblock only trusted files

A RECORDED MACRO, CLEANED UP

```
Sub FormatReport()
    ' Bold header, autofit, freeze top row, number format
    With ActiveSheet
        .Rows(1).Font.Bold = True
        .Columns("A:F").AutoFit
        .Range("D2:D1000").NumberFormat = "$#,##0.00"
    End With
    ActiveWindow.FreezePanes = False
    Range("A2").Select
    ActiveWindow.FreezePanes = True
    MsgBox "Report formatted", vbInformation
End Sub
```

CODE EXPLAINED

→ The recorder writes every scroll and select; delete lines like `ActiveWindow.ScrollRow`

→ `With … End With` saves repeating the object name

→ Lines starting with `'` are comments

→ Run from the editor with `F5`; step line by line with `F8`

OTHER SUITES

⇄ **Google Workspace** — Extensions › Macros › Record; saved as Apps Script (JavaScript)

⇄ **Apple iWork** — no macro recorder; AppleScript / Shortcuts automate iWork

PLATFORM DIFFERENCES

- **Windows** — full VBA, recorder in Word, Excel, PowerPoint (editor only), Outlook
- **Mac** — VBA works; some Windows-only objects (ActiveX, some file APIs) fail
- **Web** — VBA doesn't run; use Office Scripts (sheet 35)

VBA EDITOR KEYS

KEY	DOES
`⌥F11 / Alt+F11`	Open the editor
`F5`	Run current macro
`F8 (Mac ⇧⌘I)`	Step into, line by line
`F9`	Toggle breakpoint
`^⌘G / Ctrl+G`	Immediate window (`?Range("A1").Value`)
`⌥F8 / Alt+F8`	Macro list in Excel

WHERE MACROS LIVE

LOCATION	AVAILABLE IN
`This workbook (.xlsm)`	That file only
`Personal Macro Workbook`	Every Excel file (your PC)
`Normal.dotm`	Every Word document
`Add-in (.xlam)`	Shared with others
`Outlook VbaProject.OTM`	Outlook on that PC

⚠ COMMON MISTAKES

- Saving as `.xlsx` and losing all macros without warning (well, one warning)
- Recording with absolute references, so the macro always edits the same cells
- Enabling macros on unknown files; macros can run any code

✓ TIPS & BEST PRACTICES

- Record, then read the code; it's the fastest way to learn the object model
- Assign macros to QAT buttons for one-click access
- Add `Application.ScreenUpdating = False` at the start of long macros for speed

QUICK SUMMARY

- `Developer › Record Macro` — record steps
- `Alt+F11` — VBA editor
- `.xlsm / .docm` — macro-enabled files
- `Personal Macro Workbook` — macros everywhere

VBA Recipes

PATTERNS YOU'LL REUSE

Most useful macros are a loop over rows that checks a value and does something, an **event** that runs automatically (on open, on change), or a find-and-replace across many files. These recipes cover the object model basics: `Range`, `Cells`, `Offset`, last-row detection, events and error handling.

NOTES

- `Cells(row, col)` uses numbers; `Range("A1")` uses addresses
- Last used row: `Cells(Rows.Count, "A").End(xlUp).Row` (same as `Ctrl+↑` from the bottom)
- Events go in the sheet or ThisWorkbook module, not a regular module
- Declare variables with `Dim`; add `Option Explicit` at the top to catch typos
- `On Error GoTo Handler` sends errors to a label so the macro can clean up

LOOP, EVENT, WORD FIND

```
Option Explicit
Sub FlagOverdue()
    Dim r As Long, last As Long
    last = Cells(Rows.Count, "A").End(xlUp).Row
    For r = 2 To last
        If Cells(r, "D").Value < Date And Cells(r, "E").Value <> "Done" Then
            Cells(r, "E").Value = "Overdue"
            Rows(r).Interior.Color = RGB(255, 230, 230)
        End If
    Next r
End Sub

' Sheet module: runs whenever column B changes
Private Sub Worksheet_Change(ByVal Target As Range)
    If Not Intersect(Target, Me.Range("B:B")) Is Nothing Then _
        Target.Offset(0, 1).Value = Now      ' timestamp in column C
End Sub

Sub WordReplaceAll()                         ' Word
    With ActiveDocument.Content.Find
        .Text = "Acme Inc.": .Replacement.Text = "Acme Corp."
        .Execute Replace:=wdReplaceAll
    End With
End Sub
```

CODE EXPLAINED

- → `Long` holds row numbers past 32,767; don't use `Integer`
- → `Intersect(...) Is Nothing` checks whether the changed cell is in column B
- → `Offset(0, 1)` = one column to the right of the changed cell
- → `_` continues a statement on the next line; `:` puts two statements on one line

OTHER SUITES

- ⇄ **Google Workspace** — Apps Script: `getRange()`, `getValues()`, `onEdit(e)` trigger
- ⇄ **Apple iWork** — AppleScript `tell application "Numbers"` loops over rows

PLATFORM DIFFERENCES

- ▣ **Windows** — every recipe
- ▣ **Mac** — same code; avoid Windows paths (`C:\`) and ActiveX
- ▣ **Web** — none of this runs; port to Office Scripts

OBJECTS YOU'LL USE

OBJECT	EXAMPLE
`Workbook`	`Workbooks("Sales.xlsx")`
`Worksheet`	`Worksheets("Data")`
`Range`	`Range("A2:D50")`
`Cells`	`Cells(5, 2)` = B5
`Offset / Resize`	`Range("A1").Offset(1,0).Resize(10)`
`ActiveDocument (Word)`	`.Paragraphs(1).Range.Text`

HANDY EVENTS

EVENT	RUNS WHEN
`Workbook_Open`	File opens
`Workbook_BeforeSave`	Before every save
`Worksheet_Change`	A cell is edited
`Worksheet_SelectionChange`	Selection moves
`Document_Open (Word)`	Document opens

⚠ COMMON MISTAKES

- Changing cells inside Worksheet_Change without `Application.EnableEvents = False`, causing a loop
- Using `.Select` everywhere; act on ranges directly
- Looping cell by cell over 100,000 rows; read `Range.Value` into an array instead

✓ TIPS & BEST PRACTICES

- Use `Debug.Print` and the Immediate window to inspect values
- Turn off events, screen updating and calculation during big macros, then restore them
- Keep shared macros in an add-in (`.xlam`) instead of copying them into files

QUICK SUMMARY

- `Cells(r, c)` — by number
- `End(xlUp).Row` — last row
- `Worksheet_Change` — run on edit
- `Option Explicit` — catch typos

Sheet 35

Office Scripts & Power Automate

Automate tab · Action Recorder · TypeScript main(workbook) · run from a button · Power Automate flows · scheduled runs · VBA vs Office Scripts

AUTOMATION THAT RUNS IN THE CLOUD

Office Scripts are Excel automation written in TypeScript. They run in Excel for the web and desktop (Windows and Mac) from the **Automate** tab, can be recorded with the **Action Recorder**, and are stored in OneDrive, so they work on any machine. **Power Automate** can run a script on a schedule or when an email or form arrives, with no one opening the file.

NOTES

- Automate tab › **Record Actions** writes a script as you click
- Every script has `function main(workbook: ExcelScript.Workbook)`
- Scripts are shared through the file or a SharePoint folder
- Add a script button to the sheet: script ⋯ › **Add button**
- Power Automate: "Run script" action in a scheduled or triggered flow (business plans)

AUTOMATE IT

- Schedule a flow: every Monday 7:00, run "CleanupExport" on the latest file
- Trigger on Microsoft Forms response → add row → run formatting script

OFFICE SCRIPT (TYPESCRIPT)

```
function main(workbook: ExcelScript.Workbook) {
  const sheet = workbook.getWorksheet("Data");
  const used = sheet.getUsedRange();
  const values = used.getValues();              // 2-D array

  // Trim text in column A, flag amounts over 1000 in column E
  for (let r = 1; r < values.length; r++) {
    values[r][0] = String(values[r][0]).trim();
    values[r][4] = Number(values[r][3]) > 1000 ? "Review" : "";
  }
  used.setValues(values);

  sheet.getRange("1:1").getFormat().getFont().setBold(true);
  used.getFormat().autofitColumns();
  sheet.freezePanes.freezeRows(1);
}
```

CODE EXPLAINED

- → Read all values once with `getValues()`, change the array, write back once: fast
- → Arrays are 0-based; row 0 is the header
- → Scripts can't open other files; use Power Automate to pass data between files
- → TypeScript types like `ExcelScript.Workbook` give autocomplete in the editor

OTHER SUITES

- ⇄ **Google Workspace** — Apps Script (JavaScript) with time-driven triggers
- ⇄ **Apple iWork** — Shortcuts and AppleScript on the Mac; no cloud runs

PLATFORM DIFFERENCES

- ▣ **Web** — Automate tab, recorder, editor
- ▣ **Windows / Mac** — run and edit scripts in M365 desktop on business plans
- ▣ **Personal plans** — Office Scripts availability is limited; check the Automate tab

VBA VS OFFICE SCRIPTS

	VBA	OFFICE SCRIPTS
Language	VBA	TypeScript
Runs on web	No	Yes
Scheduled / triggered	No (Windows Task Scheduler hacks)	Power Automate
Word / PowerPoint / Outlook	Yes	Excel only
Access to files / OS	Full	Sandboxed

COMMON CALLS

CALL	DOES
`workbook.getWorksheet("x")`	Get a sheet
`sheet.getRange("A1:D10")`	Get a range
`range.getValues() / setValues()`	Read / write 2-D array
`sheet.getTable("Sales")`	Get a table
`table.addRows(-1, rows)`	Append rows
`range.getFormat().setFill…`	Formatting

⚠ COMMON MISTAKES

- Calling `getValue()` inside a loop for every cell; batch with `getValues()`
- Expecting a script to read a second workbook; that needs Power Automate
- Writing a script for a one-time cleanup that Flash Fill or Power Query would do

✓ TIPS & BEST PRACTICES

- Record first, then edit the generated code
- Return values from `main` so Power Automate can use them in later steps
- Keep scripts small and single-purpose; chain them in a flow

QUICK SUMMARY

- `Automate › Record Actions` — record a script
- `main(workbook)` — entry point
- `getValues / setValues` — batch read / write
- `Power Automate` — schedule and trigger

Copilot & Python in Excel

AI AND PYTHON INSIDE THE APPS

Copilot drafts and rewrites in Word, analyses and builds formulas in Excel, turns documents into decks in PowerPoint, and summarises threads in Outlook. The `COPILOT()` function brings AI into a cell. **Python in Excel** (`=PY()`) runs Python with pandas and charts in the cloud, returning results to the grid. Both are tools to check, not to trust blindly.

NOTES

- Copilot needs a qualifying plan or add-in; Personal / Family include some features
- Good prompts name the source, task, format and audience
- `=COPILOT("Classify this feedback as positive / negative", B2)` works row by row
- Python in Excel: type `=PY(` or Formulas › Insert Python; reference cells with `xl("A1:D100", headers=True)`
- Python runs in Microsoft's cloud; results can be values or Python objects

PROMPTS, COPILOT() AND PYTHON

```
# Copilot prompts (chat pane)
Word:  "Summarize this report in 5 bullets for an executive, plain language"
Excel: "Add a column that flags orders over $1,000 placed on weekends"
PPT:   "Create a 6-slide deck from /Q3-plan.docx with speaker notes"
Outlook: "Summarize this thread and list who owes what by when"

# COPILOT function in a cell
=COPILOT("Extract the city from this address", A2)
=COPILOT("Sentiment: positive, neutral or negative", B2:B50)

# Python in Excel (type =PY then Ctrl+Enter / ⌘↩ to run)
df = xl("Sales[#All]", headers=True)
df.groupby("Region")["Amount"].sum().sort_values(ascending=False)
```

CODE EXPLAINED

- → Point Copilot at a file with `/filename` so it uses your content, not general knowledge
- → COPILOT() results recalculate; paste as values when you need them fixed
- → In Python cells, `xl()` reads ranges or tables into pandas DataFrames
- → Python output can spill as values or stay as an object you click to inspect

OTHER SUITES

- ⇄ **Google Workspace** — Gemini side panel, Help me write, `AI()` function in Sheets
- ⇄ **Apple iWork** — Apple Intelligence Writing Tools; no formula AI or Python in Numbers

PLATFORM DIFFERENCES

- ▣ **Windows / Mac / Web** — Copilot in all three on eligible plans
- ▣ **Python in Excel** — Windows, Mac and web on eligible M365 plans; runs in the cloud

PROMPT RECIPE

PART	EXAMPLE
Goal	Draft a status update
Source	using /ProjectPlan.docx
Audience	for the leadership team
Format	5 bullets, under 120 words
Constraint	no jargon, include risks

VERIFY BEFORE USING

OUTPUT	CHECK
Formulas	Test on rows you can compute by hand
Numbers in summaries	Trace back to the source cell
Rewritten text	Facts, names, tone
Generated slides	Images, data, licensing
Python results	Row counts, filters, blanks

⚠ COMMON MISTAKES

- Pasting AI-generated numbers into reports without tracing them to the data
- Vague prompts ("make this better"); say what better means
- Using COPILOT() for things a normal formula does exactly (sums, lookups)

✓ TIPS & BEST PRACTICES

- Ask Copilot to explain a formula before you trust it
- Keep data in tables with clear headers; AI and Python both work better
- See the Agentic Coding set for general prompting techniques

QUICK SUMMARY

- `Copilot` — draft, analyse, summarise
- `/file` — ground prompts in your content
- `=COPILOT()` — AI in a cell
- `=PY()` — pandas in Excel

Quick Ref — Word, PowerPoint & Outlook

SHARED (02, 05)

ACTION	MACOS	WINDOWS
Paste Special	^⌘V	Ctrl+Alt+V
Paste as text	⌥⇧⌘V (Word)	Paste Options › Keep Text
Repeat last action	⌘Y	F4
Undo / redo	⌘Z / ⇧⌘Z	Ctrl+Z / Ctrl+Y
Copy / paste format	⇧⌘C / ⇧⌘V	Ctrl+Shift+C / V
Insert link	⌘K	Ctrl+K
Find / replace	⌘F / ^H	Ctrl+F / Ctrl+H
Spelling	F7	F7
Change case	⇧F3	Shift+F3
Save / Save As	⌘S / ⇧⌘S	Ctrl+S / F12
Collapse ribbon	⌘⌥R	Ctrl+F1
Search commands	⇧⌘/	Alt+Q
KeyTips	—	Alt
Print / PDF	⌘P	Ctrl+P

WORD (06–12)

ACTION	MACOS	WINDOWS
Heading 1 / 2 / 3	⌥⌘1 / 2 / 3	Ctrl+Alt+1 / 2 / 3
Normal style	⇧⌘N	Ctrl+Shift+N
Clear char format	^Space	Ctrl+Space
Styles pane	⌥⇧⌘S	Ctrl+Alt+Shift+S
Navigation pane	⌘F	Ctrl+F
Go To	⌥⌘G / F5	F5
Back to last edit	⇧F5	Shift+F5
Extend selection	F8	F8
Track Changes	⇧⌘E	Ctrl+Shift+E
New comment	⌥⌘A	Ctrl+Alt+M
Footnote / endnote	⌥⌘F / ⌥⌘E	Ctrl+Alt+F / D
Update fields	F9	F9
Field codes	⌥F9	Alt+F9
Word count	⇧⌘G	Ctrl+Shift+G
Show ¶ marks	⌘8	Ctrl+Shift+8

POWERPOINT EDITING (25–27)

ACTION	MACOS	WINDOWS
New slide	⇧⌘N	Ctrl+M
Duplicate	⌘D	Ctrl+D
Group / ungroup	⌥⌘G / ⇧⌥⌘G	Ctrl+G / Ctrl+Shift+G
Bigger / smaller text	⌘] / ⌘[	Ctrl+] / Ctrl+[
Copy while dragging	⌥ + drag	Ctrl + drag
Constrain drag	⇧ + drag	Shift + drag
Nudge finely	⌥ + arrow	Ctrl + arrow
Format pane	⌘1	Ctrl+1
Selection Pane	Arrange menu	Alt+F10
Outline: demote / promote	Tab / ⇧Tab	Tab / Shift+Tab

PRESENTING (28)

ACTION	MACOS	WINDOWS
Start from beginning	⇧⌘↩	F5
From current slide	⌘↩	Shift+F5
Next / previous	→ / ←	→ / ←
Go to slide n	n ↩	n Enter
Black / white screen	B / W	B / W
Pen / eraser	⌘P / ⌘E	Ctrl+P / Ctrl+E
Laser pointer	Pointer Options	Ctrl+L
Erase ink	E	E
Shortcut list	?	F1
End show	Esc	Esc

OUTLOOK MAIL (29–30)

ACTION	MACOS	WINDOWS
New message	⌘N	Ctrl+N
Reply / reply all	⌘R / ⇧⌘R	Ctrl+R / Ctrl+Shift+R
Forward	⌘J	Ctrl+F
Send	⌘↩	Ctrl+Enter
Search	⌥⌘F	Ctrl+E
Move to folder	⇧⌘M	Ctrl+Shift+V
Mark read / unread	⌘T / ⇧⌘T	Ctrl+Q / Ctrl+U
Flag	⌘'	Insert

CALENDAR & TEAMS (31–32)

ACTION	MACOS	WINDOWS
Mail / calendar	⌘1 / ⌘2	Ctrl+1 / Ctrl+2
New meeting	⌘N (Calendar)	Ctrl+Shift+Q
Day / week / month	^⌘1 / ^⌘3 / ^⌘4	Ctrl+Alt+1 / 3 / 4
Today	⌘T	Alt+Home
Teams search	⌘E	Ctrl+E
Teams mute / camera	⇧⌘M / ⇧⌘O	Ctrl+Shift+M / O
Raise hand	⇧⌘K	Ctrl+Shift+K
OneNote To Do tag	⌘1	Ctrl+1

Quick Ref — Excel Shortcuts

macOS / Windows · navigation · selection · entry · formatting · formulas · tables

NAVIGATION (13)

ACTION	MACOS	WINDOWS
Jump to data edge	⌘ + arrow	Ctrl + arrow
Go to A1	fn^←	Ctrl+Home
Last used cell	fn^→	Ctrl+End
Next / prev sheet	⌥→ / ⌥←	Ctrl+PgDn / PgUp
Screen down / up	fn↓ / fn↑	PgDn / PgUp
Go To / Special	^G	F5 / Ctrl+G
Jump to precedent	^[	Ctrl+[
Name Box	click + type	click + type
Scroll to active cell	^⌫	Ctrl+Backspace

SELECTION (13)

ACTION	MACOS	WINDOWS
Extend to edge	⇧⌘ + arrow	Ctrl+Shift + arrow
Select column	^Space	Ctrl+Space
Select row	⇧Space	Shift+Space
Current region / all	⌘A	Ctrl+A
Visible cells only	⌘⇧Z	Alt+;
Insert row / col	⌘⇧+	Ctrl+Shift++
Delete row / col	⌘-	Ctrl+-
Hide row / column	^9 / ^0	Ctrl+9 / Ctrl+0
Row differences	^\	Ctrl+\

ENTRY & FILL (14)

ACTION	MACOS	WINDOWS
Fill down / right	⌘D / ⌘R	Ctrl+D / Ctrl+R
Flash Fill	Data › Flash Fill	Ctrl+E
Fill selection	^↩	Ctrl+Enter
Today / now	^; / ⌘;	Ctrl+; / Ctrl+Shift+;
New line in cell	^⌥↩	Alt+Enter
Edit cell	F2 / ^U	F2
Pick from list	⌥↓	Alt+↓
Cancel entry	Esc	Esc
Paste values	^⌘V › Values	Ctrl+Alt+V › V
Insert note	⇧F2	Shift+F2

FORMATTING (23)

ACTION	MACOS	WINDOWS
Format Cells	⌘1	Ctrl+1
General number	^⇧~	Ctrl+Shift+~
Currency	^⇧$	Ctrl+Shift+$
Percent	^⇧%	Ctrl+Shift+%
Date	^⇧#	Ctrl+Shift+#
Number 2 dp	^⇧!	Ctrl+Shift+!
Bold / italic	⌘B / ⌘I	Ctrl+B / Ctrl+I
Outline border	⌘⌥0	Ctrl+Shift+&
Strikethrough	⇧⌘X	Ctrl+5
AutoFit column	double-click edge	Alt H O I

FORMULAS (15–21)

ACTION	MACOS	WINDOWS
Toggle $ refs	⌘T	F4
AutoSum	⌘⇧T	Alt+=
Show formulas	^ + grave	Ctrl + grave
Insert function	⇧F3	Shift+F3
Accept autocomplete	Tab	Tab
Array / Python run	⌘↩ (=PY)	Ctrl+Enter (=PY)
Recalculate	⌘= / F9	F9
Name Manager	⌘F3	Ctrl+F3

TABLES, PIVOTS & CHARTS (16, 22)

ACTION	MACOS	WINDOWS
Make table	^T	Ctrl+T
Filter on / off	⌘⇧F	Ctrl+Shift+L
Open column filter	⌥↓	Alt+↓
Group rows	⌘⇧K	Alt+Shift+→
Refresh pivot	Analyze › Refresh	Alt+F5
Refresh all	Data › Refresh All	Ctrl+Alt+F5
Chart from data	⌥F1	Alt+F1
Chart sheet	F11	F11

Sheet 39

Quick Ref — Excel Formulas

logic · math · lookup · dynamic arrays · text · date · LET / LAMBDA · errors

LOGIC & CONDITIONAL MATH (17)

FORMULA	DOES
`IF(test, yes, no)`	One of two values
`IFS(t1,v1, t2,v2, TRUE,else)`	First true test
`SWITCH(x, a,"A", b,"B", other)`	Match a value
`AND(…) / OR(…) / NOT(x)`	Combine tests
`IFNA(x, alt) / IFERROR`	Catch errors
`SUMIFS(sum, rng,crit, …)`	Conditional sum
`COUNTIFS(rng,crit, …)`	Conditional count
`AVERAGEIFS / MAXIFS / MINIFS`	Conditional stats
`SUMPRODUCT((r="W")*amt)`	Array math
`ROUND(x,2) / ROUNDUP / MROUND`	Rounding
`SUBTOTAL(109, rng)`	Sum ignoring hidden rows

LOOKUP (18)

FORMULA	DOES
`XLOOKUP(v, look, ret)`	Exact match
`XLOOKUP(v, look, ret, "none")`	With miss value
`XLOOKUP(v, look, ret, , -1)`	Next smaller (brackets)
`XLOOKUP(v, look, ret, , 0, -1)`	Last match
`XLOOKUP("*"&v&"*", look, ret, , 2)`	Contains (wildcard)
`XMATCH(v, look)`	Position
`INDEX(ret, MATCH(v, look, 0))`	Classic lookup
`VLOOKUP(v, tbl, col, FALSE)`	Legacy exact
`CHOOSECOLS(arr, 1, 3)`	Pick columns
`Table[Col] / [@Col]`	Structured refs (16)

DYNAMIC ARRAYS (19)

FORMULA	DOES
`FILTER(arr, cond, "none")`	Matching rows
`(a)*(b) / (a)+(b)`	AND / OR in FILTER
`SORT(arr, col, -1) / SORTBY`	Sort
`UNIQUE(arr)`	Distinct
`SEQUENCE(n)`	1…n
`TAKE(arr, 5) / DROP`	First / skip n
`VSTACK(a, b) / HSTACK`	Combine
`TOCOL(arr)`	Flatten
`E2#`	Whole spill range

LET, LAMBDA & SUMMARIES (21)

FORMULA	DOES
`LET(n, val, calc)`	Named steps
`LAMBDA(x, y, calc)`	Custom function
`MAP(arr, LAMBDA(x, …))`	Each item
`BYROW(arr, LAMBDA(r, …))`	Each row
`REDUCE(0, arr, LAMBDA(a,x, …))`	Accumulate
`GROUPBY(rows, vals, SUM)`	Summary table
`PIVOTBY(rows, cols, vals, SUM)`	Cross-tab
`A2:.A9999`	Trim empty rows

TEXT & REGEX (20)

FORMULA	DOES
`TEXTSPLIT(t, ",")`	Split (spills)
`TEXTBEFORE(t,"@") / TEXTAFTER`	Part of text
`TEXTJOIN(", ", TRUE, r)`	Join, skip blanks
`TRIM / PROPER / CLEAN`	Tidy text
`TEXT(n, "$#,##0")`	Format as text
`REGEXTEST(t, "^\d{5}$")`	Pattern test
`REGEXEXTRACT / REGEXREPLACE`	Pull / swap patterns
`LEN / LEFT / RIGHT / MID`	Classic text

DATE (20)

FORMULA	DOES
`TODAY() / NOW()`	Date / date + time
`EOMONTH(d, 0)`	Month end
`EOMONTH(d, -1)+1`	Month start
`EDATE(d, 3)`	+3 months
`NETWORKDAYS(a, b, hol)`	Business days
`WORKDAY(d, 10)`	10 work days out
`DATEDIF(a, b, "y")`	Whole years
`WEEKDAY(d, 2)`	1 = Mon

ERRORS (15)

ERROR	FIX
`#N/A`	No match: spaces, types
`#VALUE!`	Text in math
`#REF!`	Deleted reference
`#DIV/0!`	Divide by 0 / blank
`#NAME?`	Misspelled name
`#SPILL!`	Clear cells in the way
`#CALC!`	Empty FILTER result
`#NUM!`	Invalid number